

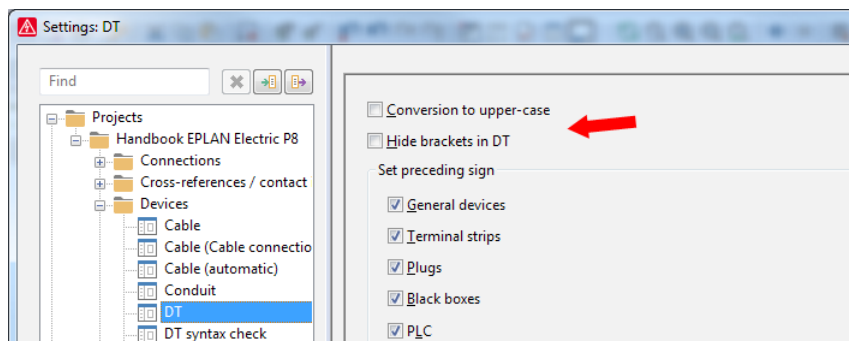


Fig. 13.43
Uppercase letters or
lowercase letters

This setting applies to the following properties in the symbols dialog:

- Displayed DT
- Full DT
- Connection point designation
- Terminal designation
- Plug DT, Channel designation, and Address (for PLC connection points)
- Structure identifiers



NOTE: If the *Conversion to uppercase* is activated/deactivated, lowercase letters will/will not be automatically converted to uppercase letters. The “conversion” only takes effect subsequent to the setting being activated/deactivated. This way, all “old” DTs entered remain in lower-case even if the setting is activated. Only the next new DT designation entered will be “converted” automatically to uppercase.

Question: How can I automatically insert a mechanical action line between two objects?

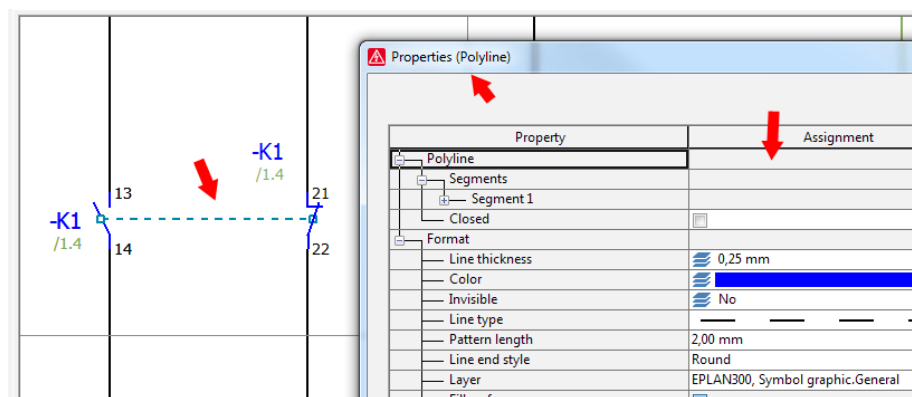


Fig. 13.44
Mechanical action line

Answer: In EPLAN, a mechanical action line is a purely graphical representation without any logical function. It is represented simply by a graphical line between two objects, or two contacts, which is then formatted according to the properties of the line.



Question: How can I open a document/image for a stored part directly from project editing?

Answer: To open a document (or image) stored for a part in parts management directly from project editing, you have to call up *properties* of the symbol and then switch to the *Parts* tab. Then, select the desired part and call the popup menu with the right mouse button.

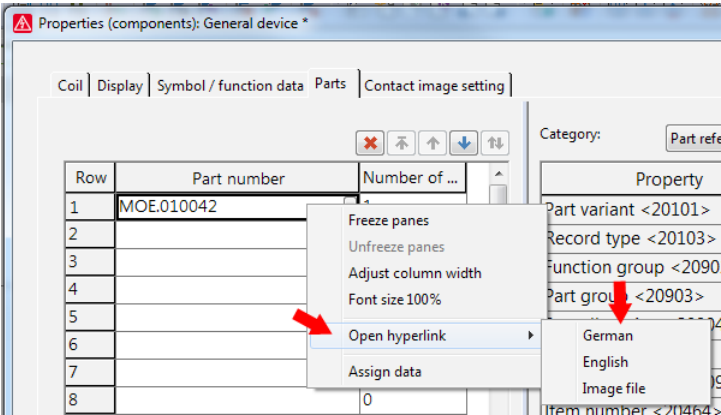


Fig. 13.45
Opening hyperlinks

The desired information can be seen in this popup menu under the **HYPERLINKS** item. Here, you can find the stored external documents 1 to n (depending on the number on the part in parts management) as well as an image of the part.

NOTE: If no such information (such as external documents or an image) is stored on the part in parts management on the *Documents* tab, the popup menu items will be grayed out and cannot be selected.



Question: What is the difference between a main function and a main function of superior device?

Answer: To understand this, you should picture a nested device. Device A (being the *main function*) is located, for example, in a black box B (also a main function). This makes black box B the *superior main function* of device A.



Question: What are the differences between *Save additionally*, *File off* and *Archive* in the data backup dialog?

Answer:

Save additionally: When backing up data, a complete (depending on the options selected) copy of the project is saved to another storage medium. In project management, it is also possible to back up multiple projects. Multiple selection is only possible in project management.

File off: If a project is to be passed on (e.g. to a customer), you can use the *File off* option. This creates a copy of the project (e.g. on a different storage medium), and the source project is set to read-only. This prevents changes in the meantime to this (filed off) project. If an attempt should still be made to edit such a filed off project, the user doing this will receive a message that the project has been filed off and locked for editing.

Archive: Finished (completed) projects can be archived in order to free up “space” on the hard disk. A copy of the project is saved to a different storage medium (e.g. the local hard disk), and the (finished) source project is deleted from the hard disk except for an information file.

Question: How can I start an Excel table via the external programs?

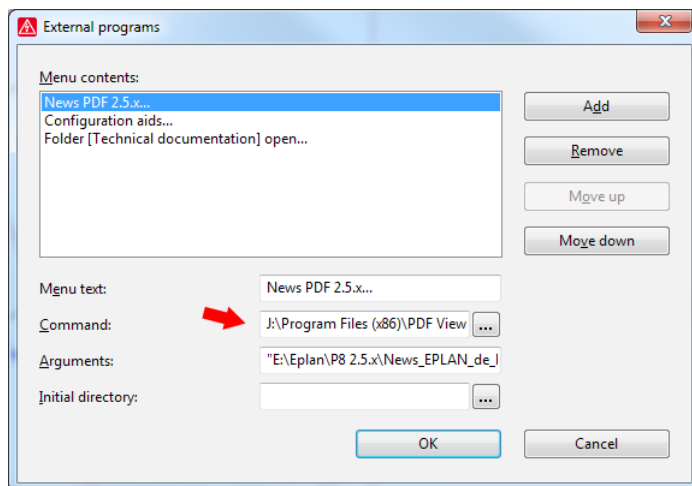


Fig. 13.46
Using external
programs

Answer:

1. Open the external programs via the **OPTIONS/EXTERNAL PROGRAMS** menu.
2. Use the **ADD** button to generate a new menu item.
3. Enter a suitable (subsequently accessible via the **OPTIONS** menu) *menu text*, for example “Design aid...”.
4. In the *Command* field, enter the path to the *Excel.exe* (for example: *J:\Program Files\Microsoft Office\Office14\EXCEL.EXE*).
5. In the *Arguments* field, enter the path to the Excel table: If there are blank spaces in the path name, then set it in quotes (for example: “F:\Excel\Color codes for Eplan E5, P8.xls”)

Save the changes with **OK**. In the **OPTIONS** menu, the “Color codes for EPLAN and others...” menu item is immediately available and opens the Excel table if it is selected.

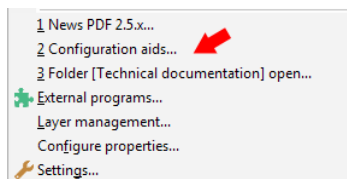


Fig. 13.47
New menu item in
the Options menu



TIP: This menu item can also be assigned its own shortcut key.

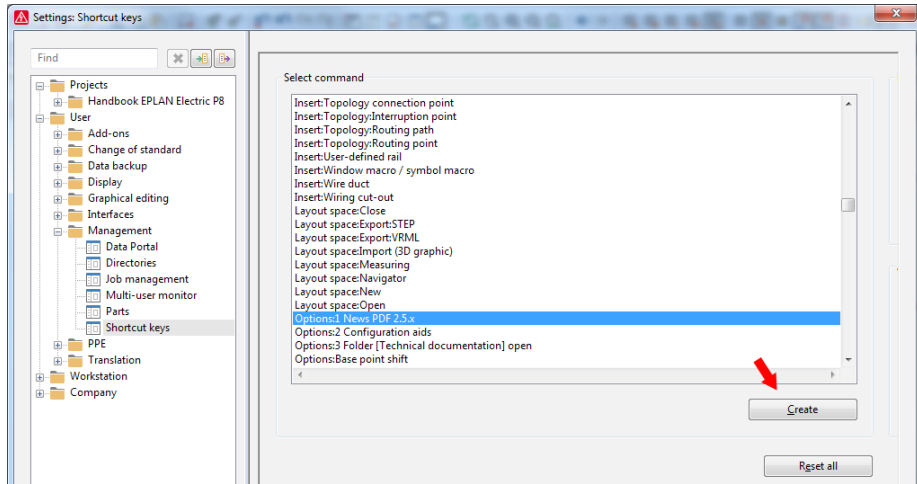


Fig. 13.48 Calling menu item with user-defined shortcut key



Question: Why is the unit of the cross-section/diameter of a connection definition not displayed?

Answer: If the unit of the cross-section/diameter is not displayed, then the wrong property arrangement has been applied. The unit is not shown by default. A different property must be set to display the unit.

To display “mm²” together with the cross-section information, you would need a property like *Cross-section/diameter with unit <31007>*. You need to add this property and format it on the *Display* tab of the **Properties (...)** dialog.



TIP: You can also manually set or selected the unit in the **Properties (...)** dialog, but this is not recommended.

YOU can also use the project default settings to define the *Unit* setting. Go to **CONNECTIONS / PROPERTIES** on the *Electrical engineering* tab and fill in the *Unit* field. This setting applies globally and is the recommended method.

Question: Is it possible to add information to the display window on the right in the **Open project** dialog?

Answer: No. This information cannot be extended and is permanently defined by EPLAN.

Question: Where and how can I define a description for my structure identifiers?

Answer: To be able to sort structure identifiers or other descriptions, you have to enter them directly in **STRUCTURE IDENTIFIER MANAGEMENT** (PROJECT DATA menu).

Question: Can I change the row height (and thus the font size of the display), for example in the **Properties** (components) dialog or in the 'Edit in table' mode, etc., in order to be able to read it better?

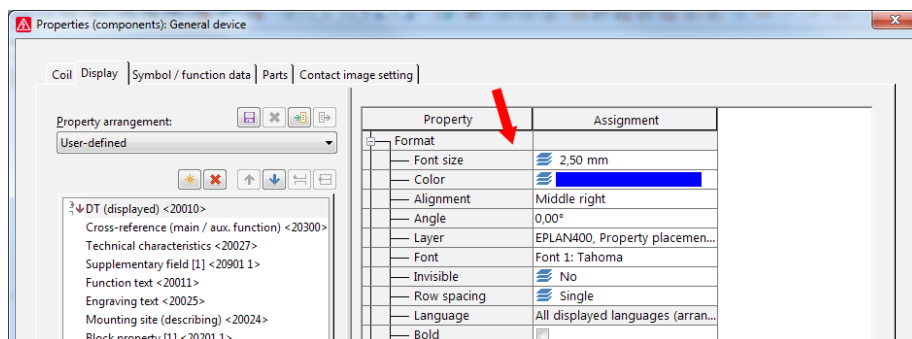


Fig. 13.49 Changing the row height – before

Answer: Yes, this is possible. To increase the row height (and thus the font size of the display), you click the relevant display, keep the **CTRL** key pressed and turn the scroll wheel of the mouse. Depending on the direction, the display will be increased or reduced.

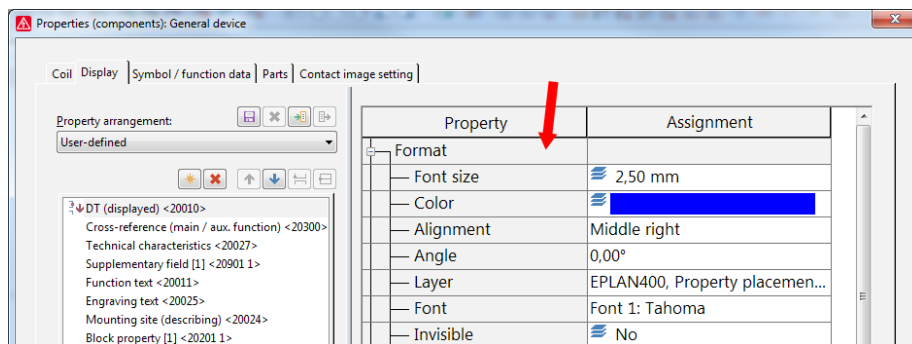


Fig. 13.50 Changing the row height – after



Question: Can I assign the *Find new DT* function in the **Properties (components) ...** dialog in the *Displayed DT* field (right mouse button) to a shortcut key?

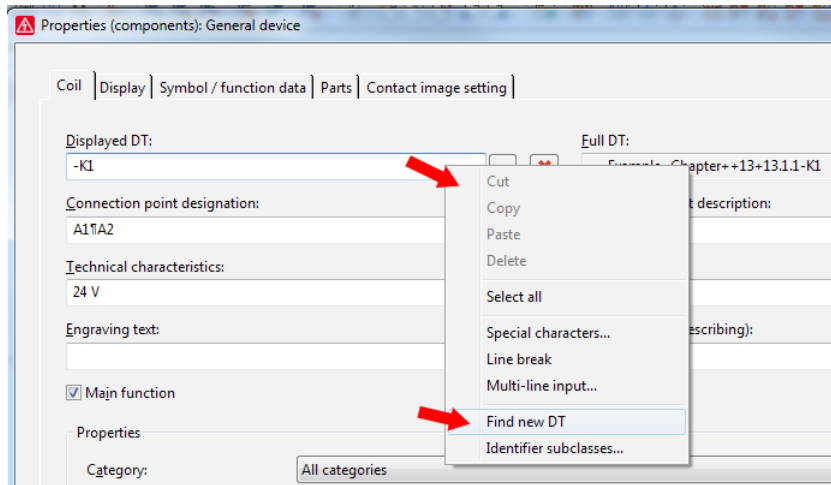


Fig. 13.51 Find new DT

Answer: You cannot use your own shortcut keys, but if you put the cursor in the *Displayed DT* field and press **CTRL + N** shortcut key (you must keep the **CTRL** key pressed), EPLAN will execute the function shown in the popup menu (**FIND NEW DT**).



Question: I received a check run message that is not relevant to me.

Answer: If checks are not necessary, you can deactivate them in the test scheme. Open the test scheme for editing, locate the corresponding check run messages and set the type of check to "No".



Question: Is there a way to fully reset the manual page sorting in the page navigator?

Answer: Yes, this is possible. You must take the following steps to fully reset the manual page sorting to default sorting:

1. Open the **page navigator** and select *List view* (tab at the bottom end of the page navigator).
2. Click the right mouse button while in the page navigator and select **MANUAL PAGE SORTING**.
3. Again right click in the dialog and select **MANUAL PAGE SORTING**.

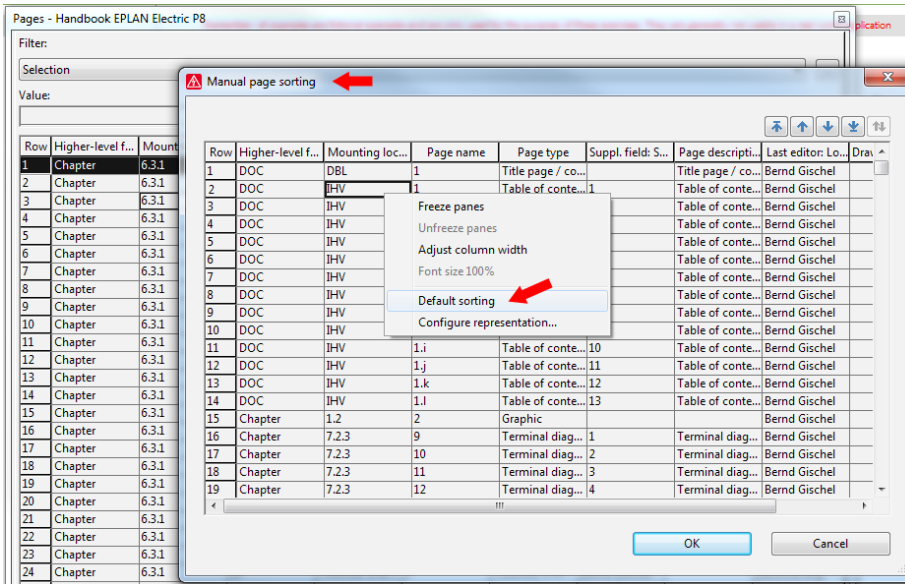


Fig. 13.52 Resetting manual page sorting

EPLAN resets manual page sorting in the page navigator to the default sorting.

Question: Is it possible to automatically find objects placed outside the plot frames?

Answer: Yes. EPLAN has a *check run message* for this. This must be activated in order for EPLAN to find objects placed outside the plot frame automatically. It is found in the check run settings under *ID 010 Cross-references* and has the message number “010001” with the message text “The placement lies outside the evaluation range of the plot frame”.



NOTE: The check run message captures only logical objects. Free objects like texts, etc. are not captured by this check.

13.2 Parts

Question: While selecting a device, I noticed my part had a typing error in Designation 1. Can I remove this typing error directly without having to start parts management?

Answer: Yes, this is possible. Under **USER/MANAGEMENT/PARTS MANAGEMENT**, *Modification allowed during selection* must be activated. Then, you can call the part directly on the symbol and modify it.

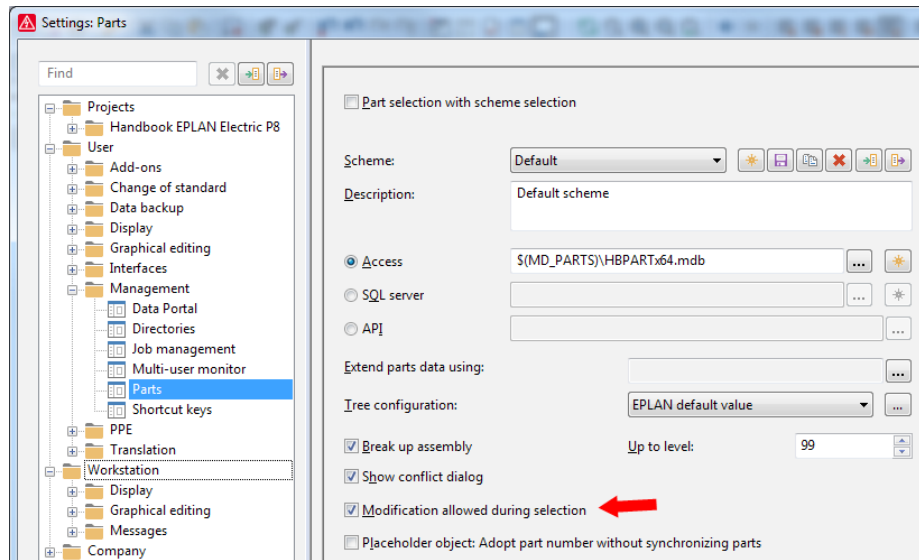


Fig. 13.53 Changing parts data setting during selection



Question: Is it possible to synchronize the stored parts data automatically upon opening the project?

Answer: Yes, this is possible. The *Synchronize stored parts when opening* setting in the project settings must be activated.

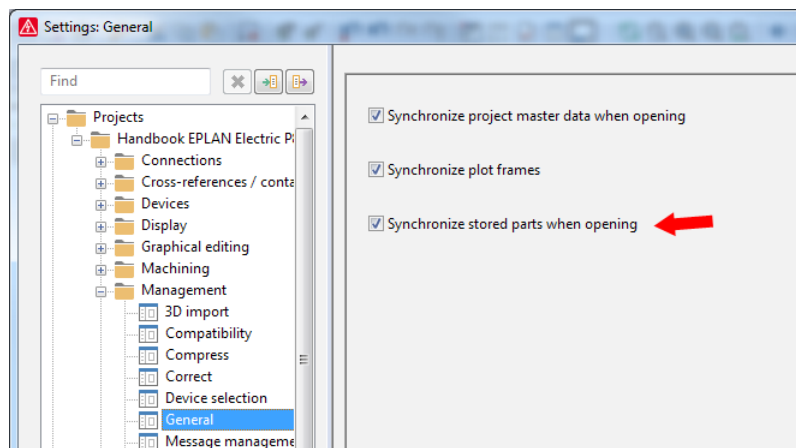


Fig. 13.54 Updating stored parts when opening the project



Question: Can I store more than one document on a part in parts management?

Answer: Yes, absolutely. EPLAN allows you to store up to 20 documents on a part in parts management. To do so, they must be entered in parts management for the desired part on the *Documents* tab and/or selected from the directories and imported.

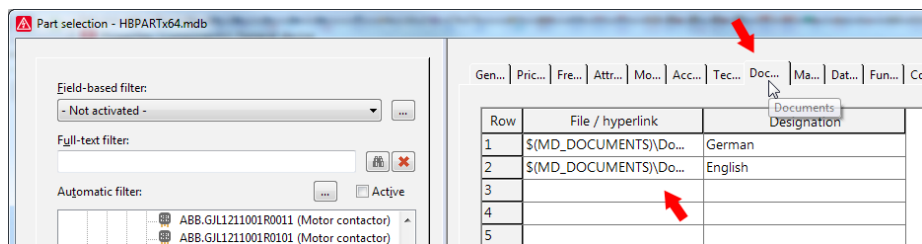


Fig. 13.55 Documents tab

Question: Why can I not enter a part on my function (the Parts tab is missing)?

Answer: Only *main functions* can have parts data. A function must be a main function in order to support parts data. If this feature is missing (the *Main function* setting has not been checked, e.g. on the *Symbol/function data* tab), the *Parts* tab will not be available either.

Question: What is the difference between device selection and part selection?

Answer: In the case of *part selection*, EPLAN does not check whether the part “matches” the device in the project when the part on the device is selected via the More button [...] for the *Part number* on the *Parts* tab. To put it simply, you could assign a PLC card to a motor overload switch.

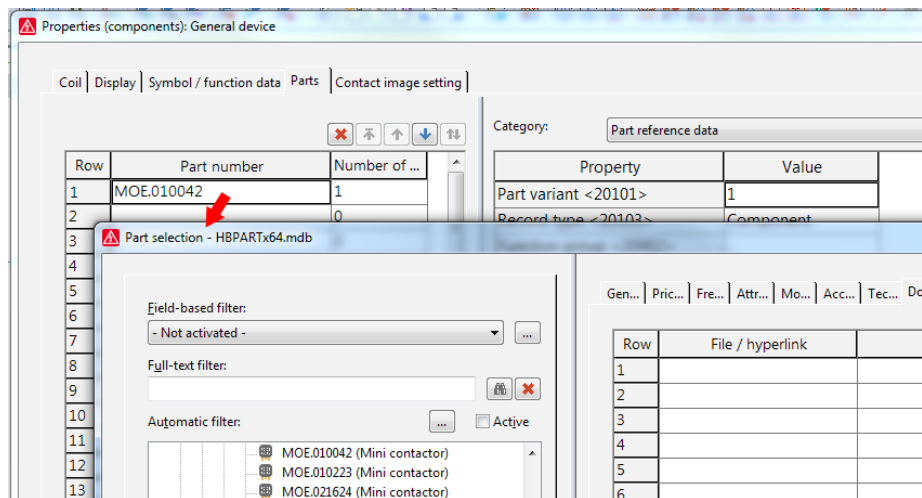


Fig. 13.56 Simple part selection

In the case of *device selection*, when you click the device selection button EPLAN automatically checks which device functions already exist and only displays parts (devices) that match the function definitions. That is to say, with device selection, it is not possible to assign to a motor overload switch the PLC card device.

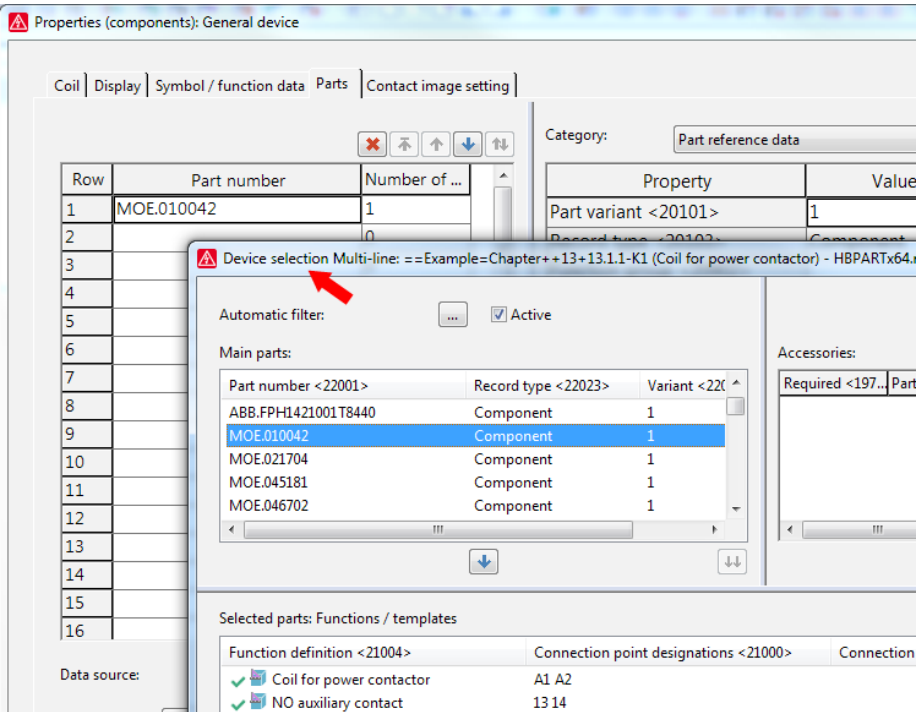


Fig. 13.57
Device selection



Question: How can I export all (or individual) parts in the project to a new parts management?

Answer: To export project parts used (stored) to a new parts management, you have to first create a new database in parts management. Open parts management and click EXTRAS. Here, you select the SETTINGS function. EPLAN opens the **Settings: Part (User)** dialog. A new database is generated here via the *New* button.

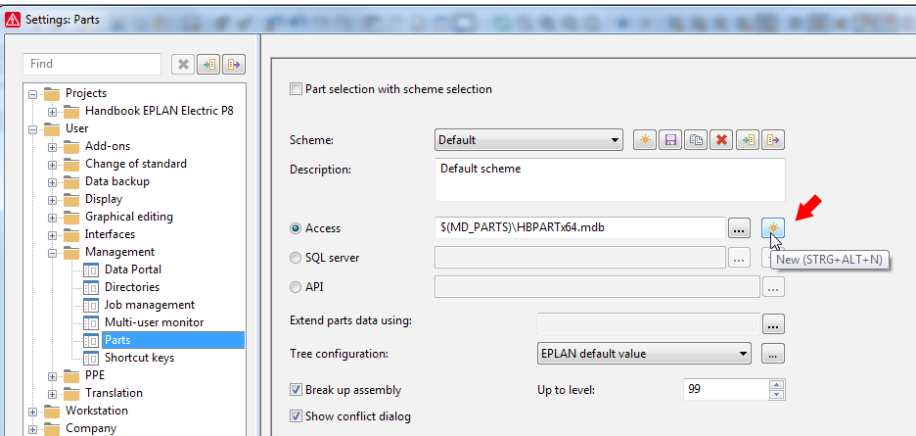


Fig. 13.58
Creating a new
database

Click the *Open* button in the **New database** dialog, and EPLAN will apply the new database and enter it in the settings as the current database. Now leave the **Settings** dialog and parts management (by clicking OK and/or clicking the *Close* button in parts management).

In the UTILITIES/PARTS menu, select the SYNCHRONIZE CURRENT PROJECT function. EPLAN opens the **Synchronization of parts** dialog. The filter in the dialog is set to *- Not activated -*. EPLAN now lists all project parts. You can select them all, and use the button to move them to the new database on the right.

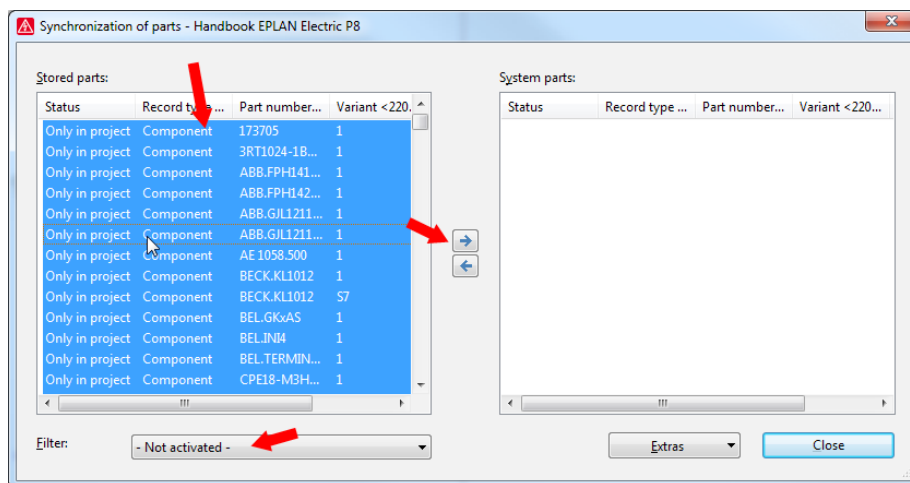


Fig. 13.59 Storing project parts in the new database

EPLAN starts synchronization without further prompts. This way, all project parts are stored in a separate parts database.

Question: In parts management, how do I create new product groups, generic product groups or subgroups?

Answer: For reasons related to the data exchange, this is generally impossible. New generic product groups, product groups as well as new product subgroups cannot be created in the EPLAN parts management by the user. These groups are fixed and defined by EPLAN.

Question: In assemblies, how can I enter the function templates of individual devices?

Answer: You do not have to do this manually. EPLAN does this automatically. Simply click the EXTRAS button in parts management and select the SUM UP FUNCTION TEMPLATES menu item. EPLAN now collects the function templates of the individual devices and automatically enters them in the assembly's *Function template* tab.





Question: In parts management, is it possible to copy an existing function template of a part?

Answer: Yes, this is possible. To transfer (copy) a suitable function template of an existing part to another part that does not have a function template, you must do the following:

1. Start parts management.
2. Select a part with a suitable function template in the tree or list view.
3. Right click to open the popup menu and select the **COPY** function.
4. Now select the desired part (the one without a function template) in the tree or list view.
5. Right click to open the popup menu and select the **PASTE FUNCTION TEMPLATES** function.
6. EPLAN now transfers the function template to the part.



NOTE: If the part already has a function template, a confirmation prompt before pasting will ask you whether the existing function template should be overwritten or not. Make sure to confirm this.



Question: Is it possible to create from stored parts data a device list for an external project?

Answer: Yes, this is possible. To do so, you must export the parts data using a suitable labeling scheme and then import them in the external project as a device list.



Question: How can I translate the *Designation 1* text of parts?

Answer: To translate a part's *Designation 1* text, proceed as follows:

1. Start parts management.
2. Click the *Designation 1* field of the desired part.
3. Right click to open the popup menu and select **TRANSLATE**.
4. When EPLAN opens the **Found words** dialog, select the correct translation.
5. Click OK to apply the translation.

The *Designation 1 text* has now been translated and can be used in multilingual forms and other reports.



NOTE: The following condition must be met prior to any translation: The text to be translated must already exist in the dictionary.

Question: How can I make sure that parts to be provided with delivery are not listed in the reports?

Answer: To prevent, for example, parts supplied by the customer from appearing in reports, all you need to do is to give this part a characteristic that you can use later on for filtering purposes.

You could, for example, use the *Part group* in the part reference data for this characteristic. You could enter CS for customer-supplied or any other input here. You can later use this characteristic to filter out these parts so that they do not appear in reports.



TIP: In parts management on the *Technical data* tab, you can preset the *Part group* field to CS for parts that are generally customer-supplied. This way, the *Part group* is filled automatically with the entry CS on the device when the part is assigned.

Question: Where is a project-specific parts database stored?

Answer: The project-specific parts database is stored in the project directory `...\*.edb` under the file name *part.mdb*.

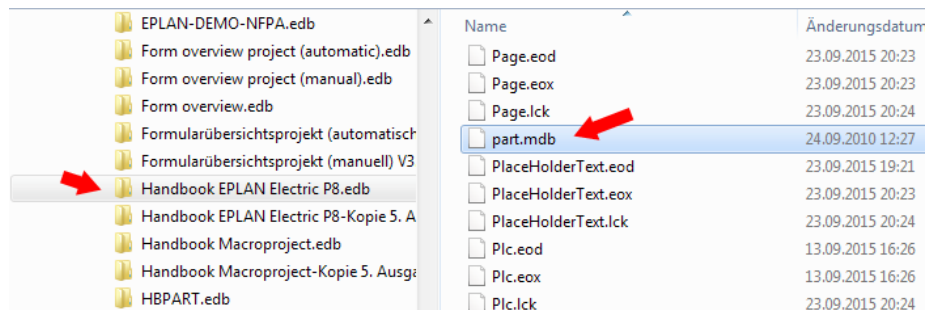


Fig. 13.60 Project-specific parts database

Question: There are many parts missing from my parts management. Where can I obtain the desired parts?

Answer: EPLAN provides only a limited selection of various parts (as sample parts) in the parts database *ESS_part001.mdb*. There are several ways to obtain additional parts. You can use the EPLAN Data Portal or visit different manufacturers' websites to look for EPLAN parts data. Finally, though, you can and will have to create the missing and/or desired parts in parts management yourself.

13.3 Terminals, plugs



Question: Can one and the same terminal strip be output in two different reports?

Answer: No, this does not work (unless you freeze the first report). Since terminal strips are function-specific reports, there is no way in EPLAN to output one and the same terminal strip twice in a terminal diagram. Function-specific reports may and can occur in a project only once.



Question: Why do the terminal strip parts entered not appear in the parts list?

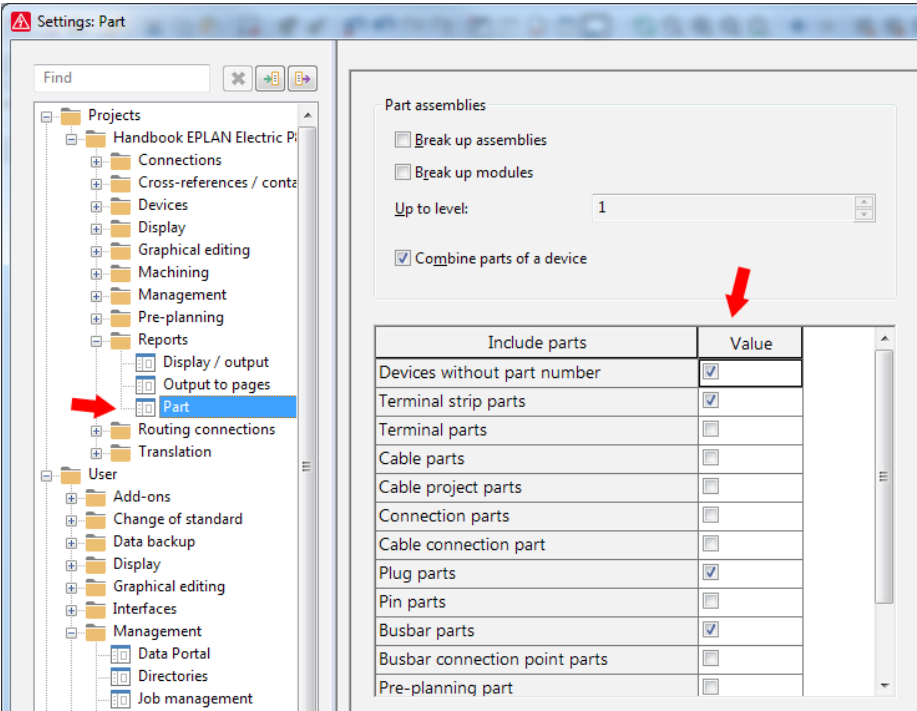


Fig. 13.61 Include parts

Answer: If articles like *terminal strip parts*, *terminal parts*, *cable parts*, *cable project parts*, *cable conductor parts* *connection parts*, etc. are to be included in reports, for example, in the parts list, you must set the check mark in the **VALUE** column for the relevant parts (to be included) in the report settings under *Include parts*. The same is true of the desired option to output a parts list with all devices, even if they do not have any parts data (Device without part number).



Question: How can I assign “superior” function text to a terminal strip?

Answer: To assign function text to a terminal strip (for example, so that a report can be created on it in a terminal-strip overview), you must enter the function text at the terminal strip definition.

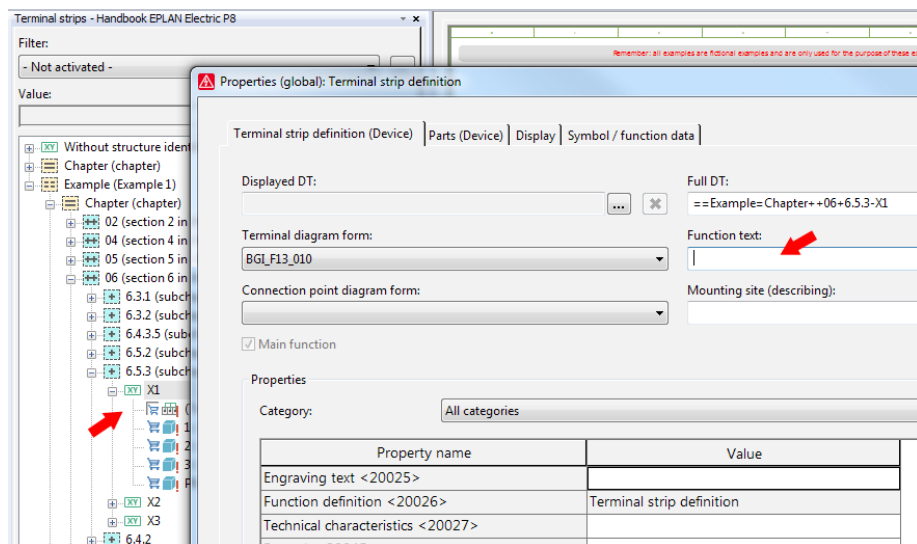


Fig. 13.62 Function text on terminal strip definition

If the terminal strip definitions do not exist, under **PROJECT/ORGANIZE/CORRECT** you can use an appropriate schema to have the terminal strip definitions be generated automatically by EPLAN.

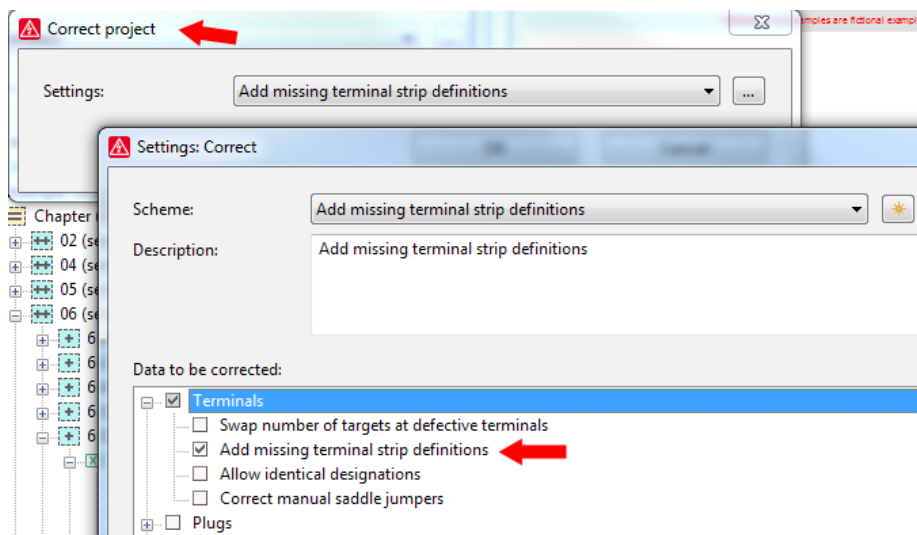


Fig. 13.63 Correcting terminal strip

 **NOTE:** For the correction function to work, you do not have to select any terminal in the project. EPLAN automatically looks for the correct and missing objects, and then adds them all to the project. An individual selection is not possible.



Question: How can I generate plug definitions automatically?
Answer: To generate plug definitions automatically, go to the **PROJECT/ORGANIZE/CORRECT** menu to generate an appropriate scheme using the activated *Add missing plug definition* option.



Question: How is it possible for terminals to have the same terminal designation, without EPLAN outputting an error message?
Answer: To avoid error messages in the check run for terminals sharing the same terminal designation, e.g. in the case of PE terminals and the PE terminal designation, you must activate the *Allow same designations <20811>* property in the symbol properties of the terminal.

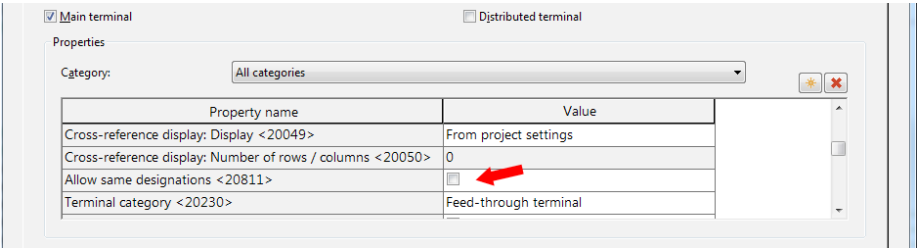


Fig. 13.64
Allow same terminal designations property



Question: How can I remove superfluous terminal strip or plug definitions from the project automatically?
Answer: To avoid having to delete superfluous terminal strip definitions (or plug definitions) from the terminal strip or plug navigator manually, you create a scheme in the **PROJECT/ORGANIZE/COMPRESS** menu that will remove this superfluous data from the project automatically.

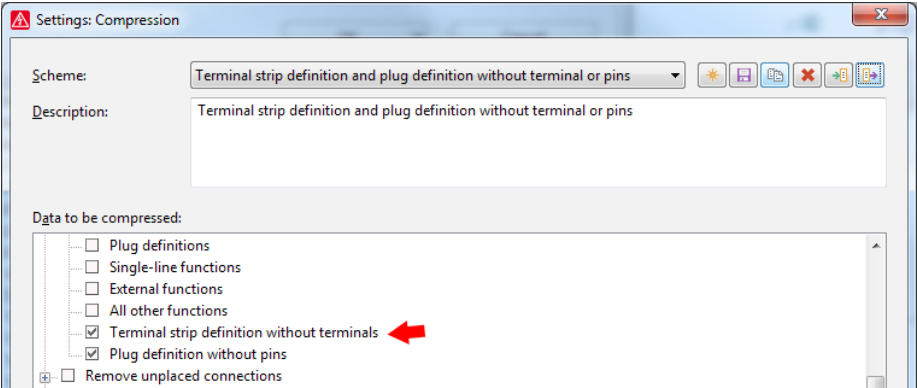


Fig. 13.65
Compression scheme

Question: How can several terminal diagrams be output on a report page in a space-saving manner?



Answer: To output several terminal diagrams (terminal strips) on a single report page, several conditions must be met. The form to be used must be a dynamic form. As well, in the **Reports** dialog (UTILITIES/REPORTS/GENERATE menu) under the **SETTING/OUTPUT TO PAGES** button, a check mark must be placed for the respective report in the *Combine* column.

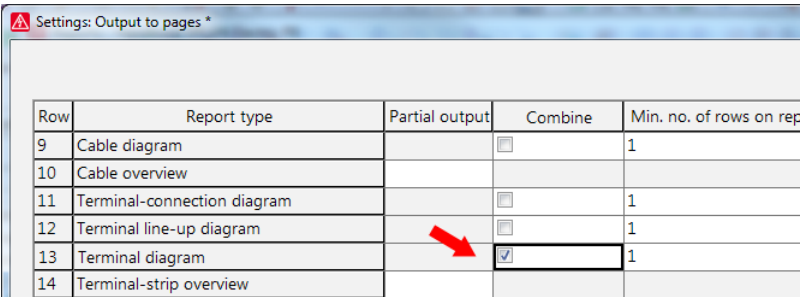


Fig. 13.66 Activate combination for a report

In addition to the *terminal diagram* report, the *Combine* option is also available for the following reports (sorted by the F number):

- Device connection diagram *.f05;
- Cable diagram *.f09;
- Cable connection diagram *.f07;
- Terminal diagram *.f13;
- Terminal connection diagram *.f11;
- Terminal line-up diagram *.f12;
- Enclosure legend *.f18;
- PLC diagram *.f19;
- Plug diagram *.f22;
- Pin connection diagram *.f21;
- Symbol overview *.f25;
- Mounting list *.f32;
- Topology: Routing path diagram *.f35
- Pre-planning: Structure segment plan *.f39
- Pre-planning: Planning object plan *.f41
- Pre-planning: Segment template plan *.f43
- Assembly/Module overview *.f44
- Distributed device list *.f45
- Bundle/conduit plan *.f46

■ 13.4 Cables



Question: Why can't EPLAN execute a correct device selection of cables (incl. assignment of conductor information) on open interruption points?

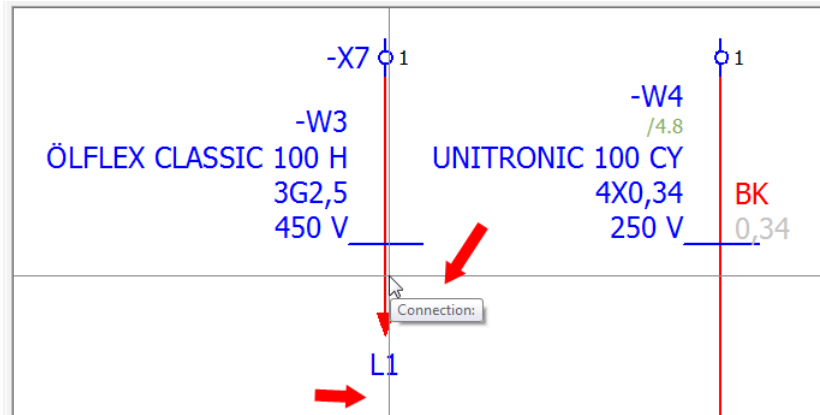


Fig. 13.67 Open connection

Answer: An open interruption point in EPLAN is not automatically considered a logical connection. This can be seen very well when you place the cursor on the Autoconnecting line and then view the tool tip. This is why device selection and the automatic placement of conductors is supported only on valid (connected) logical connections.



Question: What are the identifying properties of a cable definition?

Answer: The following properties of a cable are identifying properties: The actual *function definition*, the *shield designation* (Shielded by), the *pair index*, the *potential type*, the (conductor) *color* and/or *number* and the *cross-section* and/or *diameter*.

Accordingly, these properties must match the function definitions of a device selection when the cable and/or its connection data were previously edited manually (e.g. the cable conductors or the pair index may have been inserted manually).

If one of these properties does not match the cable (in case of a device selection) in parts management, the device selection will not be able to find a suitable cable.



Question: Why are cable details, e.g. conductors of a cable, listed multiple times in the cable navigator?

Answer: If cable details such as conductors or the shopping cart are listed in the cable navigator multiple times, it indicates most likely that several cable definitions (or lines) (displayed in a distributed manner) exist in the project and that they are all marked as *main functions* containing one part each.

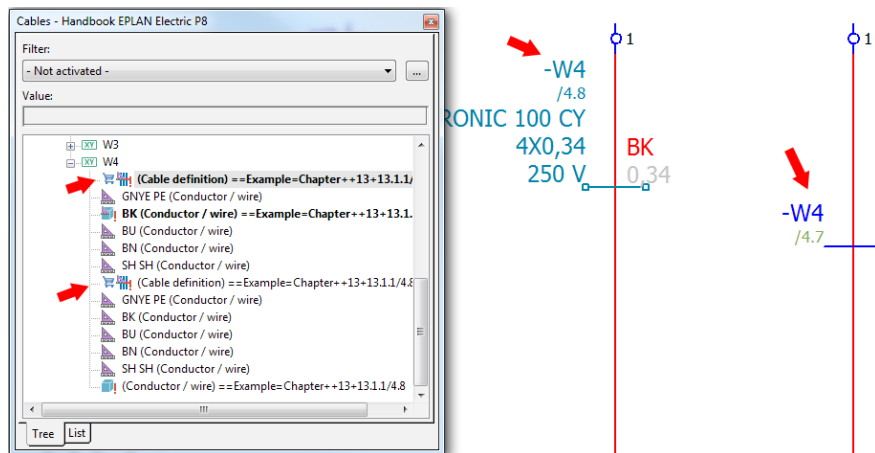


Fig. 13.68

Duplicate cable definition in the navigator

In the case of cables, too, EPLAN only allows for one main function. However, it does not matter to EPLAN where, with which representation type or on which cable definition the main function and the corresponding parts data are located.

Question: How can the conductors of a cable be assigned automatically in EPLAN?

Answer: To have EPLAN automatically “fill” in the conductor information for a cable (like color or cross-section), you need to create the cable as a part complete with all conductors in parts management.

If, when inputting the cable in the schematic, you use device selection to select and apply the cable, then when you exit the **Symbol properties cable** dialog, the conductors (crossing the connections) will be labeled by EPLAN automatically based on the parts management data.

Question: How does EPLAN define the source and target of a cable connection?

Answer: When creating reports, EPLAN goes from right to left and from bottom to top and then determines the source and target. This means that when the cable runs from left to right, then the source would be on the left and the target on the right. The same is true of bottom to top. In this case, the source would be at the bottom and the target at the top.

13.5 Properties, layers

Question: How can I carry over the property arrangements quickly from one device to another?

Answer: To transfer the painstakingly created arrangements of properties from one device to another, you can use the **COPY FORMAT** and **ASSIGN FORMAT** functions in the **EDIT** menu in EPLAN.

How is a property arrangement transferred? First select the (completely) formatted device. Then go to the **EDIT** menu and select the **COPY FORMAT** command. Then select the device to which the previously copied format is to be transferred. Now go to the **EDIT** menu and select the **ASSIGN FORMAT** command. The format of the property arrangement has been transferred to the device.



TIP: Block functions such as multiple selection can be used here. You can select multiple devices on a page and simultaneously transfer the form to them. This also works in the navigators. In this case, you need to select the devices to be modified and assign the format.



NOTE: Formats can be transferred only to similar (same function) devices. You could not, for example, transfer a cable format to a motor since it has a totally different function.



Question: On which layer is the “anchor” (placeholder object) for a macro with value sets located?

Answer: The layer of the so-called anchor (placeholder object; symbol 323/PLHO) of a macro with value sets is located on *EPLAN322 Symbol graphic.Macro.Placeholder object*. Here, for example, you can make settings whether the symbol is to be printed or whether it is to be visible in the diagram.

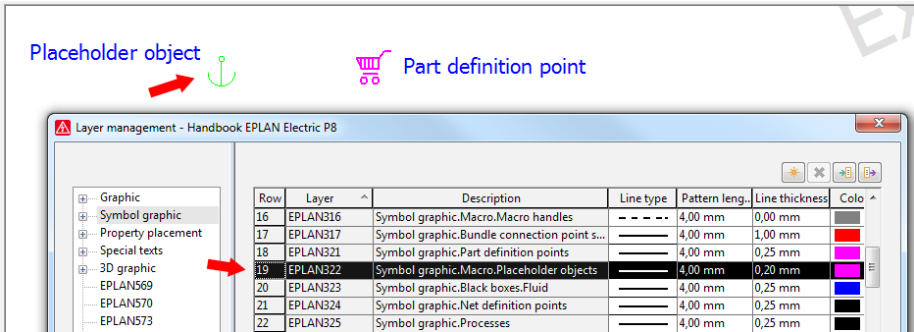


Fig. 13.69
Properties of the EPLAN322 layer

The symbol with the symbol number 323 (symbol name PLHO) originates from the SPECIAL symbol library and cannot be modified, because the SPECIAL symbol library is protected against modifications by EPLAN.



Question: On which layer is the part definition point symbol (symbol number 80, symbol name PDP) located?

Answer: The part definition point symbol (**INSERT – PART DEFINITION POINT** menu) is located on the layer *EPLAN321 Symbol graphic.Part definition points*.

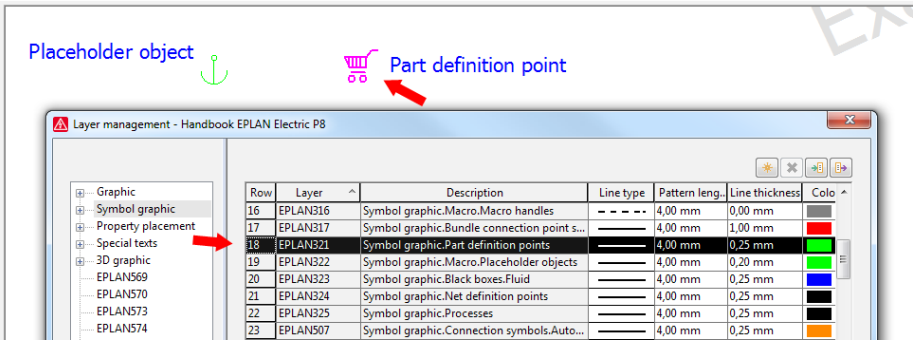


Fig. 13.70
Properties of the
EPLAN321 layer

The symbol with symbol number 80 (symbol name ADP) is also from the SPECIAL symbol library. This symbol cannot be changed since the SPECIAL symbol library is write-protected by EPLAN.

Question: How can I change the thickness (line thickness) of a dimension line (dimensioning functions)?

Answer: To change the thickness of a dimension line for different dimensioning functions, you can/must edit the *EPLAN107 (Graphic.Dimensions)* layer in layer management.

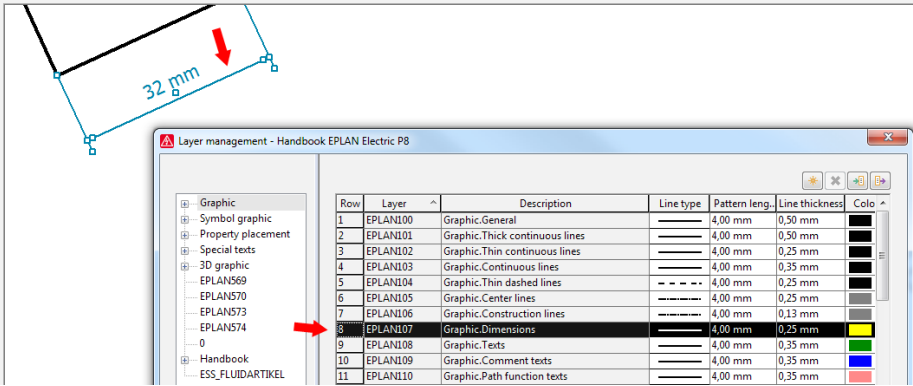


Fig. 13.71
Properties of the
EPLAN107 layer

Question: What is the meaning of the three different options for the *cross-reference display <20021> property*?

Answer: There are three options for the *Cross-reference display <20021> property*.

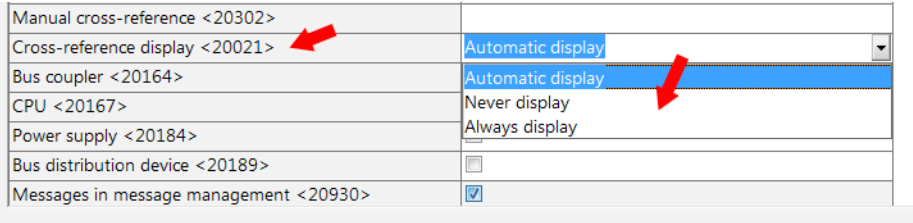


Fig. 13.72
Differences in cross-
reference display
<20021>

They regulate the display of automatic cross-references between the main function and its auxiliary functions and function as follows:

Automatic display: A cross-reference is displayed only if the *Displayed DT* field is filled with a correct DT, i.e. when it is not empty.

Never display: A cross-reference is never displayed; existing cross-references are suppressed in the display.

Always display: Cross-references are generally displayed.

Independent of the displays, there is always a logical connection between the main function and the auxiliary functions. All navigation commands, such as *GO TO...*, work as usual.



Question: Where can I change the text size of the part number or the type number above/below the contact image?

Answer: This information (formatting options) is located on the layer *EPLAN480 Property placement.Part number* (for the part number) and on the layer *EPLAN481 Property placement.Type number* (for type designation).

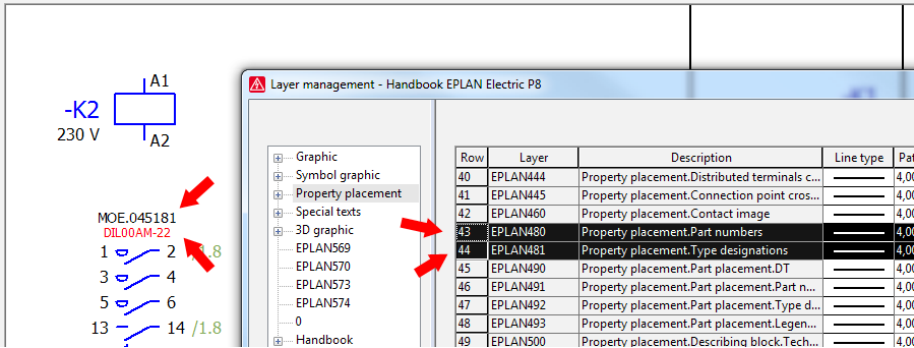


Fig. 13.73 Properties of the layers EPLAN480 and EPLAN481



Question: Do the layer management settings generally apply globally, or are they project-specific?

Answer: The layer management settings (called via the *OPTIONS/LAYER MANAGEMENT* menu) always refer to the current project.

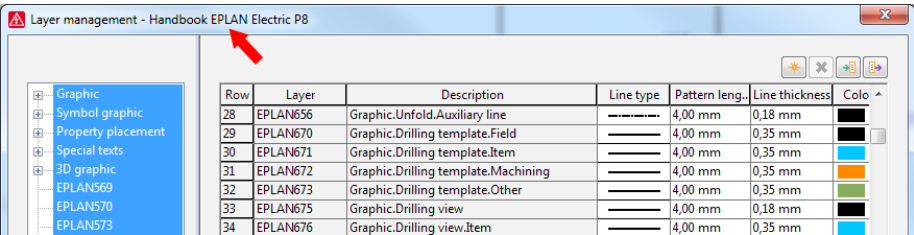


Fig. 13.74 Layer management and the project

Question: On which layer are the color settings for e. g. the alignment box of texts located?

Answer: The alignment box, for example, can be adjusted on the *Format* tab in the text dialog. It does not have a separate layer of its own. All settings (color, invisible, etc.) are controlled via the *EPLAN108 Graphic.Texts* or *EPLAN110 Graphic.Path function texts*-layers.

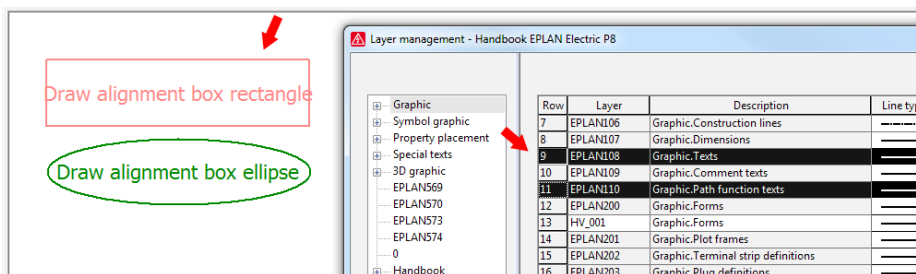


Fig. 13.75 Properties of the layers EPLAN108 and EPLAN110

Question: How can I display the description of structure identifiers on devices?

Answer: If you wish to display the descriptions of structure identifiers on devices, you simply have to show on the desired device the corresponding property in the display of properties (*Display – Property arrangements – Components* tab).

The following properties are possible, for example, in order to display structure identifier descriptions:

- *Description: Higher-level function* <1130>
- *Description: Higher-level function number* <1730>
- *Description: Installation site* <1430>
- *Description: User-defined* <1630>
- *Description: Document type* <1530>
- *Description: Mounting location* <1230>
- *Description: Functional assignment* <1330>



13.5.1 Master data

Question: How can I include row numbers in the plot frame?

Answer: You must enter the following plot frame settings in the plot frame editor:

- First, open the plot frame via the **UTILITIES/MASTER DATA/PLOT FRAME/OPEN** menu to edit it.
- Insert the *row texts* in the graphical editor with the desired number (**INSERT/SPECIAL TEXT** menu).



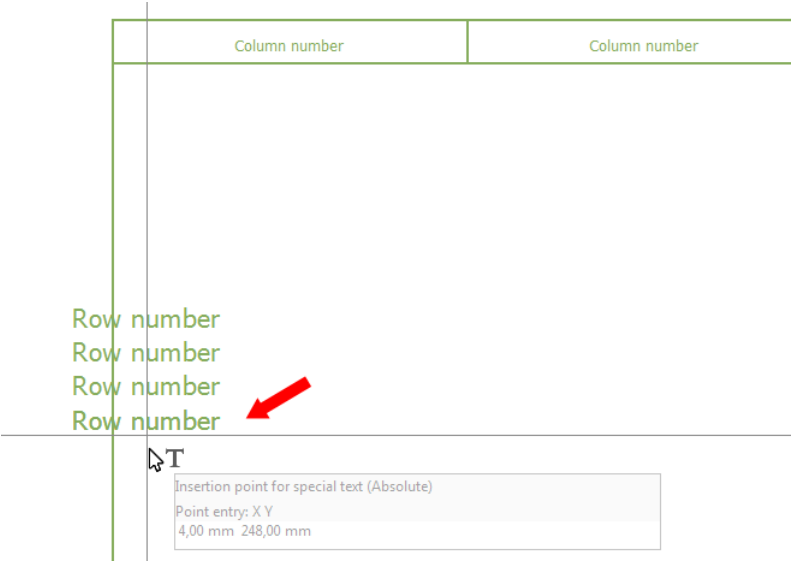


Fig. 13.76 Inserting row texts

- Use **CTRL + M** followed by **CTRL + D** to call the *Plot frame properties*.
- Enter the *number of rows*, define the *row height*, the *strings* and the *numbering format* for the rows and exit the plot frame properties by clicking **OK**.

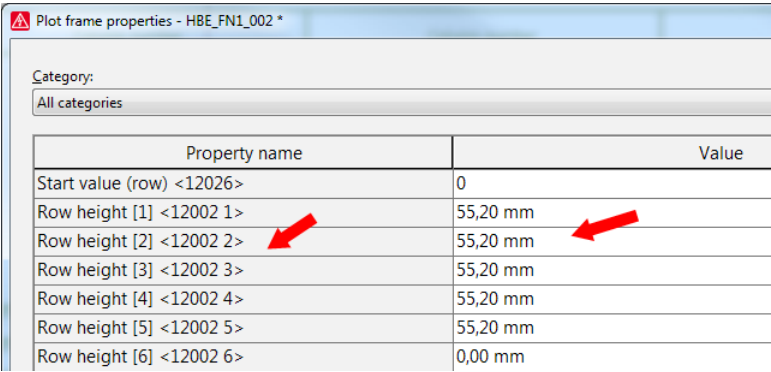


Fig. 13.77 Entering properties for the rows

- Select **UTILITIES/PLACE COLUMN AND ROW TEXT AGAIN**. EPLAN now automatically places the *column texts* and *row texts* anew on the basis of the widths and heights taken from the plot frame properties. If necessary, you can manually correct these distances or enter a graphical line if you want.

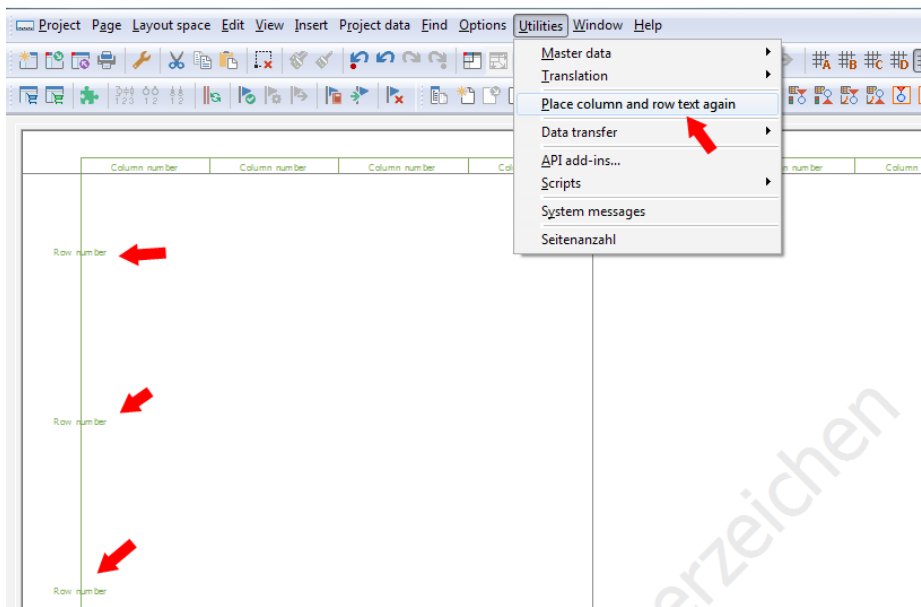


Fig. 13.78 Letting EPLAN determine the distances automatically

Editing of the plot frame is complete. It can now be closed and used.

Question: What is the difference between the symbols XBD and XBD2?

Answer: Given the correct “counter symbols”, one combination does not generate a connection (XBD), while the other does (XBD2). This way, as well, the counterpart of the plug (female pin) is found and cross-referenced.

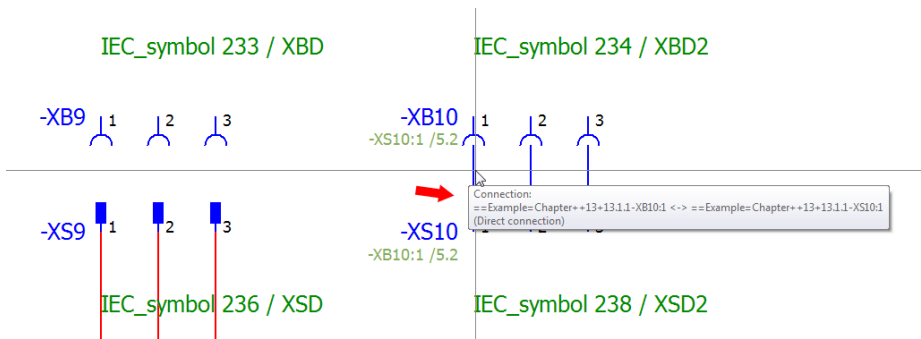


Fig. 13.79 Differences between XBD/XBD2 and XSD/XSD2 symbols

Question: Why is my report form, e.g. a terminal diagram, larger than my plot frame?

Answer: This is an imported EPLAN 5 project. The plot frame (a holdover from the old EPLAN 5 version) was left in the project settings, but reports are now created on the basis of the EPLAN Electric P8 report forms. Since plot frames imported from the old EPLAN 5



Question: How can I assign to a form a separate plot frame as default, so that the standard plot frame is not imported from the project settings?

Answer: To assign to a report (a terminal diagram for example) generally a plot frame different from the standard plot frame of the project, you must do the following.

- Open the desired report (form) via **UTILITIES/MASTER DATA/FORM/OPEN**.
- Select and open the desired form in the **Open form** dialog.
- In the *Form properties* (accessible by pressing **CTRL + M** shortcut key followed by **CTRL + D**), set the *Use 'Plot frame to edit form' property for reports <13055>* property and then set the desired plot frame in the *Plot frame to edit form <13001>* property that should be automatically set in the future for this report (form).

Property name	Value
Plot frame to edit form <13001>	HBE_FN1_005
Use 'Plot frame to edit form' property for reports <13055>	☒
Form handling <13002>	Static
Adapt row height dynamically <13102>	☐

Fig. 13.81
Customizing form properties

- Close the form properties and then the form.

From now on, this form will always be automatically assigned the plot frame defined in the form properties for the purposes of graphical output.

Question: What form information is shown in the **Open form** dialog on the right-hand side, and is it possible to add to this information?

Answer: The following form information appears in the right-hand field of the **Open form** dialog.

- Description: Description (form, plot frame, outline) <18011>
- Company code:
- Creator: Bernd Gischel
- Creation date: (Format) Day.Month.Year
- Last edited by: Bernd Gischel
- Modification date: Format Day.Month.Year
- Form handling: Static form

Description: Terminal diagram
 DIN A3 Landscape
 2 Columns
 26 Rows
 Based on DIN_KLP_32_2_dyn
 Company code:
 Creator: mka
 Creation date: 18.04.2005
 Last editor: SIL

Fig. 13.82
Form properties in the
Open form dialog

This information shown cannot be extended.



Question: What form information is shown in the **Open plot frame** dialog on the right-hand side, and is it possible to add to this information?

Answer: The following form information appears in the right-hand field of the **Open plot frame** dialog.

- Description: Description (form, plot frame, outline) <18011>
- Company code:
- Creator: Bernd Gischel
- Creation date: (Format) Day.Month.Year
- Last edited by: Bernd Gischel
- Modification date: (Format) Day.Month.Year

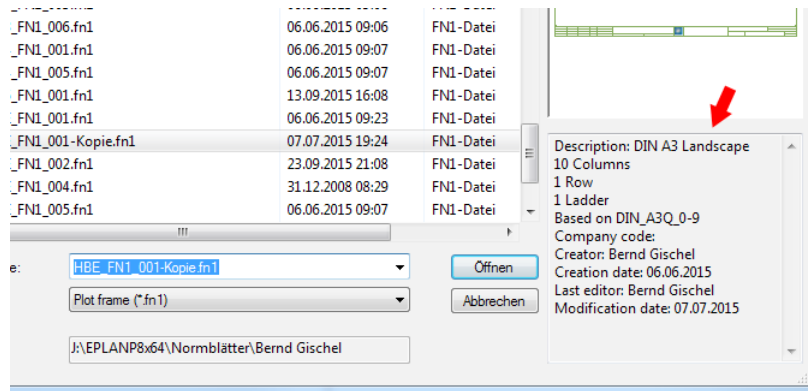


Fig. 13.83 Plot frame properties in the Open plot frame dialog

Here, too, it is not possible to add further information to this display.



Question: How can I edit a form that is located only in the project received from a supplier and not in the system?

Answer: To edit a form (report) that is located only in the project master data and not in the system master data – e.g. a project sent by an external supplier who uses proprietary forms in the project – you must do the following:

1. Open the project that contains the form.
2. In the **UTILITIES** menu, select **MASTER DATA/UPDATE CURRENT PROJECT**.
The **Synchronize master data** dialog opens.
3. Select the form not stored in the system master data in the left area of the dialog (the project master data) and use the button in the center to copy it to the EPLAN system master data on to the right.
4. Close the dialog. Now, the form (report) can be edited as usual in EPLAN (master data editor).



NOTE: This approach also applies to the remaining master data only stored in the project, such as plot frames, etc.

Question: How can I change the “fixed” project properties, e.g. on the cover/title page?

Answer: You can open the project properties directly via the **PROJECT/PROPERTIES** menu path and change the desired entries in the *Properties* tab.

Or you could select the project in the **page navigator** (you only need to select one page), call up the popup menu via the right mouse button and then also select **PROJECT/PROPERTIES**.

Question: When I generate a report, I receive the message “S079005 form does not exist.” What can I do?

Answer: If EPLAN outputs this message when you generate a project report or a specific report, then you are missing the relevant form that was entered in the page outputs or report template.

In the **Reports** dialog, under the **SETTINGS** button, open the **OUTPUT IN PAGE** menu item. Here, select an existing form in the Report row, e.g. terminal diagram, from the *Form* column. Or you can select and apply an existing form in the relevant (report) template (in the **Reports** dialog; *Templates* tab) from the *Form* row.

Question: How can I set a symbol (or its symbol graphic) to invisible?

Answer: It is not possible to set a symbol (or the actual symbol graphic) to invisible. To set a symbol (or the actual symbol graphic) to invisible, the existing symbol must be exchanged for a symbol from the symbol libraries that precisely matches this symbol, with the exception that its symbol graphic properties are set to “invisible”. If there is no “invisible” (counter) symbol in the symbol libraries, the symbol cannot be set to invisible.

Question: Is it possible to “convert” symbols to graphic, for example, to reduce or enlarge them for internal connections?

Answer: Yes, this is possible, and it is quite easy to do. You simply click the symbol to select it and then call up the **EDIT/OTHER/CONVERT COMPONENT INTO GRAPHIC** menu. EPLAN then “breaks down” the symbol and generates a separate handle for each element.

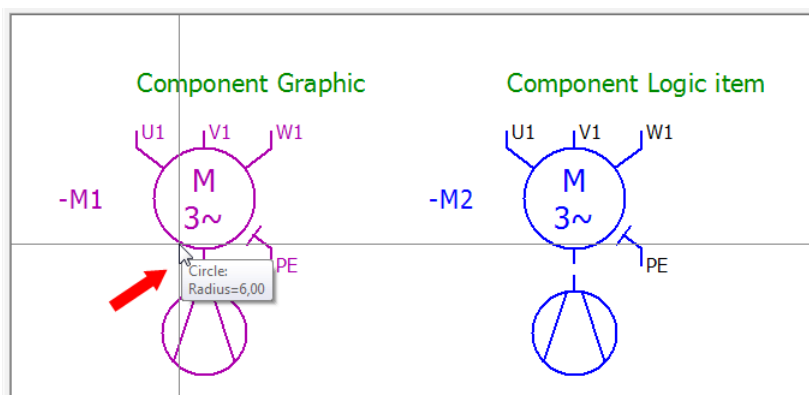


Fig. 13.84 Component converted into graphic



NOTE: This action can only be reversed immediately (i.e. as soon after the conversion as possible). If the project has been closed, or the **UNDO** list has been cleared for any other reason, you will no longer be able to “undo” the conversion (**GRAPHIC > COMPONENT / SYMBOL**).



Question: How do I remove a symbol library from the project?

Answer: Generally, this only works when all symbols in use from this symbol library have been removed from the project completely. Otherwise EPLAN will not let you remove the symbol library because these symbols might still be in use.

■ 13.6 Data exchange



Question: Does EPLAN Electric P8 have an EXF interface like the previous EPLAN 5?

Answer: No, EPLAN Electric P8 does not have such an EXF interface. It isn't needed anymore because existing interfaces to programs like Excel allow you to easily and conveniently change data in EPLAN Electric P8 “externally”.



Question: How do I convert EPLAN Electric P8 projects to the previous EPLAN 5 or EPLAN 21 versions?

Answer: Logically (i.e. with evaluable information), this is not possible at all. EPLAN Electric P8 projects are not backward compatible with EPLAN 5 or EPLAN 21. In purely graphical terms, for example in the DXF/DWG exchange format, this can be done. Of course, this may be subject to certain limitations regarding representation, line thickness or font size.



Question: Is there a viewer for EPLAN Electric P8?

Answer: Yes. There is a viewer for EPLAN Electric P8. It can be purchased from EPLAN.



Question: Should imported EPLAN 5 or EPLAN 21 projects be used as a basis for new projects in EPLAN Electric P8?

Answer: Although the import function is quite good (special or totally exotic “distorted” legacy projects aside), I personally would not use imported EPLAN 5 or EPLAN 21 projects as a basis for future EPLAN Electric P8 projects.

Of course EPLAN 5 or EPLAN 21 projects could be enhanced to include the new EPLAN Electric P8 options, and at first this appears to require less effort than creating a new EPLAN Electric P8 project from scratch. But converted data can be tricky or not really be optimized for the new system (in this case, EPLAN Electric P8).

An example would be the additional auxiliary contacts of a motor overload switch, which in EPLAN 5 had to be drawn on the actual motor overload switch to obtain a cross-reference from the pair cross-reference. This approach is also possible in EPLAN Electric P8, but not necessary. It is not necessary, because a motor overload switch with auxiliary contacts can be created as a part including the function template, thus displaying the “contact image” on the motor overload switch automatically following device selection. This means that you do not have to set auxiliary contacts again, only to display the cross-reference to the auxiliary contact used on the motor overload switch (and its auxiliary contacts).

There are a number of examples where EPLAN Electric P8 has substantially more elegant and more convenient solutions. This is why starting from scratch in EPLAN Electric P8 is the better approach.



NOTE: EPLAN 21 projects can no longer be imported directly in EPLAN Electric P8 (version 2.4 and higher). Therefore, an older version (prior to version 2.4) must be used.

Question: How can I export specific device information to Excel?

Answer: To easily export device information (properties) to Excel, you should use the **Labeling** module. The **Labeling** module is located in the **UTILITIES/REPORTS** menu. Generally, there are several file formats to choose from: Excel format **.xls*, text file format **.txt* and XML file format **.xml*.

EPLAN provides a series of complete *schemes* that can be used to export specific properties of certain devices, such as cables, terminals or general devices, to Excel. These schemes can be copied. The properties to be exported can then be modified on the *Label* tab. But it is also possible to create your own, new schemes on the basis of the various *report types* like terminal diagram, device tag list, etc.

The scheme (for exporting to Excel) contains many format output options on the various tabs (*Header*, *Footer* and *Label*). You can also set up filter and sorting options in the schemes (**Settings** tab).

Question: Can I use programs other than Microsoft Excel to export and import EPLAN Electric P8 data?

Answer: No, this is not possible. Other program packages, such as OpenOffice, etc., cannot be used for functions such as the export and import of labeling or for external edit functions. This data exchange generally requires Microsoft Excel.



13.7 Reports



Question: Is it possible to generate a report for a device tag list with the graphical overview of complex devices, for example, as demanded by energy supply companies?

Answer: Yes, EPLAN can do that. One possibility is to use conditional forms.

You must do the following in EPLAN:

1. Generate a graphical overview of the device. Ideally, you should create a separate symbol library where such overviews can be created and saved.

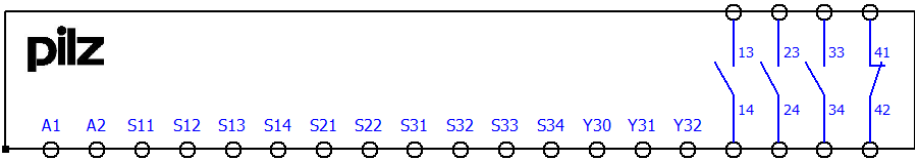


Fig. 13.85 Graphical overview

2. On the *Data for reports* tab in parts management, assign this symbol to the part for which a report with this graphical overview is to be generated in the device tag list.

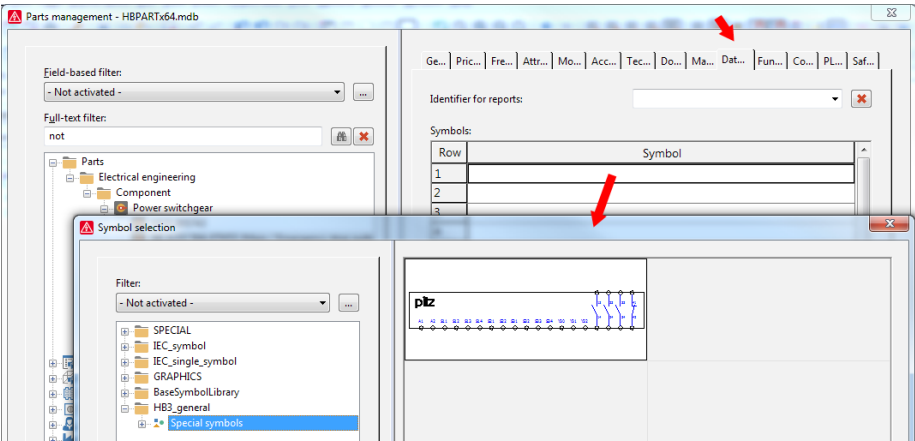


Fig. 13.86 Assign symbols to part

3. Then, you need to create corresponding device tag list forms, so that the representation can later be displayed on them as well. You can start with a basic form that displays all other devices without detailed representation as before.

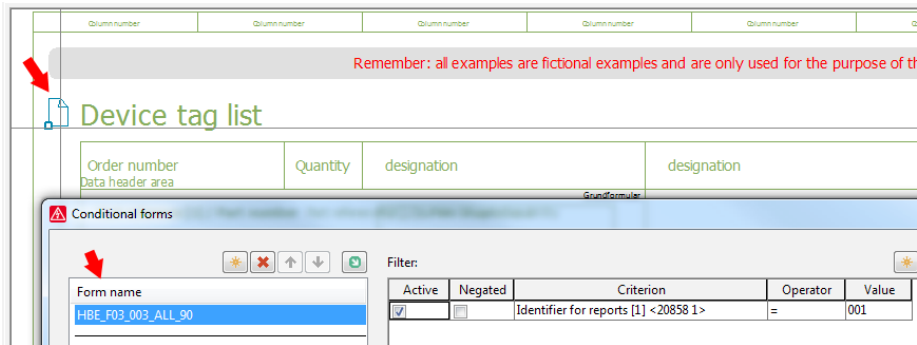


Fig. 13.87 Basic form editing

4. Into this standard form you place a conditional form (via the **INSERT/CONDITIONAL FORMS** menu) and use the  button in the **Conditional forms** dialog to select the subform for the graphical detailed representation.

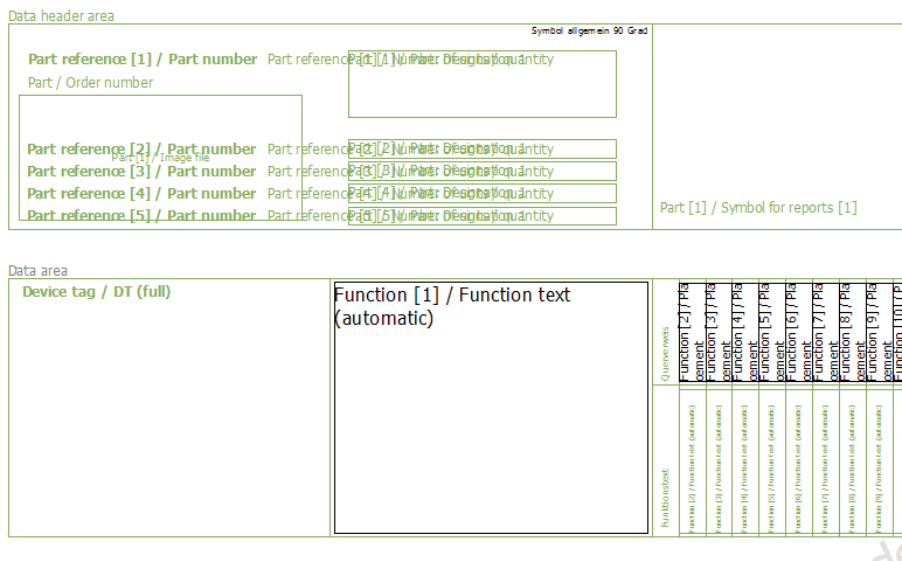


Fig. 13.88 Subform

5. The subform can be called conveniently and directly from the basic form for editing. Once the subform is closed again, EPLAN returns to the **Conditional forms** dialog. In this dialog, it is important to note that a *code* is now defined for the report, so that EPLAN, when generating a report, can correctly assign the part (with its code on the *Data for reports* tab) and the *code* of the conditional form (subform).
6. If all these preconditions have been met, you can generate a report of the device tag list. EPLAN generates the report and displays the devices accordingly.

re fictional examples and are only used for the purpose of these exercises. They are generally not usable in a real world application

HBE_F03_003_G

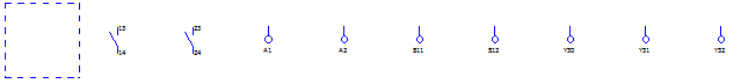
	designation
switching device	
	/B.1 /B.5 /B.7 /B.2 /B.2 /B.2 /B.2 /B.2 /B.2 /B.3

Fig. 13.89 Standard report without detailed representation

Device tag list

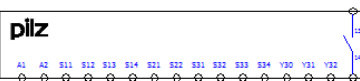
Order number	Quantity	designation	designation																												
Symbol allgemein 90 Grad																															
MUS.NOTAUS4 NOTAUS4	1	EMERGENCY STOP safety switching device																													
-A1	#####																														
		#####	<table><tr><td>Querschnitt</td><td>/B.2</td></tr><tr><td>Querschnitt</td><td>/B.2</td></tr><tr><td>Querschnitt</td><td>/B.2</td></tr><tr><td>Querschnitt</td><td>/B.2</td></tr><tr><td>Querschnitt</td><td>/B.2</td></tr><tr><td>Querschnitt</td><td>/B.2</td></tr><tr><td>Querschnitt</td><td>/B.3</td></tr><tr><td>Querschnitt</td><td>/B.3</td></tr><tr><td>Querschnitt</td><td>/B.3</td></tr><tr><td>Querschnitt</td><td>/B.3</td></tr><tr><td>Querschnitt</td><td>/B.4</td></tr><tr><td>Querschnitt</td><td>/B.2</td></tr><tr><td>Querschnitt</td><td>/B.2</td></tr><tr><td>Querschnitt</td><td>/B.4</td></tr></table>	Querschnitt	/B.2	Querschnitt	/B.2	Querschnitt	/B.2	Querschnitt	/B.2	Querschnitt	/B.2	Querschnitt	/B.2	Querschnitt	/B.3	Querschnitt	/B.3	Querschnitt	/B.3	Querschnitt	/B.3	Querschnitt	/B.4	Querschnitt	/B.2	Querschnitt	/B.2	Querschnitt	/B.4
Querschnitt	/B.2																														
Querschnitt	/B.2																														
Querschnitt	/B.2																														
Querschnitt	/B.2																														
Querschnitt	/B.2																														
Querschnitt	/B.2																														
Querschnitt	/B.3																														
Querschnitt	/B.3																														
Querschnitt	/B.3																														
Querschnitt	/B.3																														
Querschnitt	/B.4																														
Querschnitt	/B.2																														
Querschnitt	/B.2																														
Querschnitt	/B.4																														

Fig. 13.90 Detailed report with symbol and special report

This “emergency stop combination” is, of course, a relatively simple example. It is also possible and conceivable to think of examples of complex devices with internal schematics or plug-in design, etc.

14

Creating a schematic project – step by step

Experience has shown that difficulties can arise at certain “editing points” in EPLAN Electric P8 even though the solution is fairly apparent. This chapter will take a step by step look at a few practical examples to help eliminate such stumbling blocks.



NOTE: In EPLAN Electric P8, there is not “one right” way to do something. When you go through the examples in this chapter, always remember the same result could be achieved in another way.

To use EPLAN, you first need to use an existing project template to create a schematic project that has a specific structure. Project planning in EPLAN roughly consists of the following steps:

1. Create a project
2. Generate pages
3. Create a schematic
4. Generate reports



■ 14.1 Create a project

What is a project? In EPLAN, schematics, terminal diagrams, etc. are created and generated as pages in a project. Simply put, a project is a large database where all the parts of a project are stored.

What is a project structure? In EPLAN, a project structure is the structure that should be used for the project. Structures are identifiers, such as a higher-level function (=) or the installation site (++). All objects in the project are stored in such a project structure and sorted within this structure.

What is a project template? EPLAN needs a project template in order to create a new project. A project template defines certain settings, though they usually can be changed later.

EPLAN offers many ways to create a new project. In this example, we call up the **PROJECT/NEW** menu.

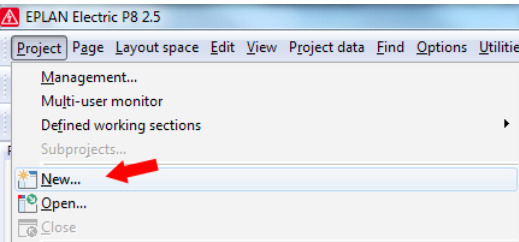


Fig. 14.1
Project/New menu

EPLAN opens the **CREATE PROJECT** dialog shown in Fig. 14.2 where settings need to be made.

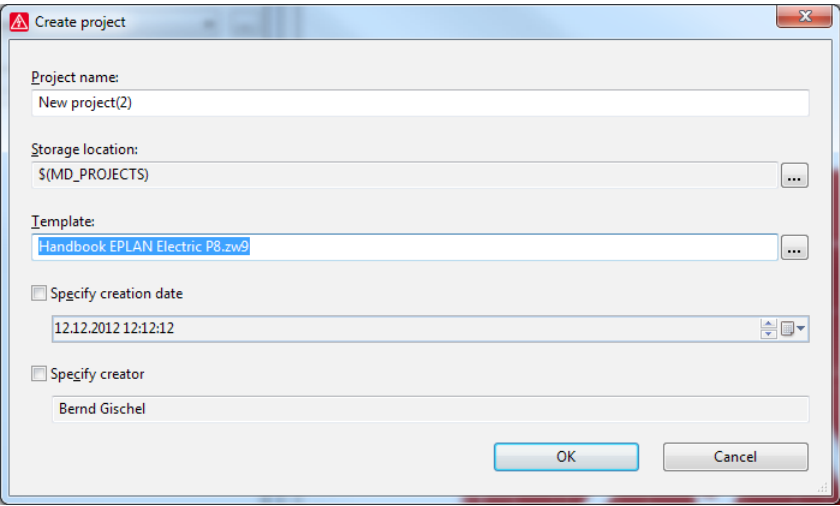


Fig. 14.2
Create project dialog

Table 14.1 The most important inputs for a new project

Setting	Comment	Example
Project name	Enter the name of the project here.	Sample project
Storage location	Use the [...] button to select the location where the project should be stored.	E:\EPLANP8\Projects
Template	Use the [...] button to select the template that the project should be based on.	IEC_bas002.zw9
Creation date	Enter the desired date and time.	27.09.2015 xx:xx:xx
Creator	Enter the name of the person who is to be listed as the creator of the project.	Bernd Gischel

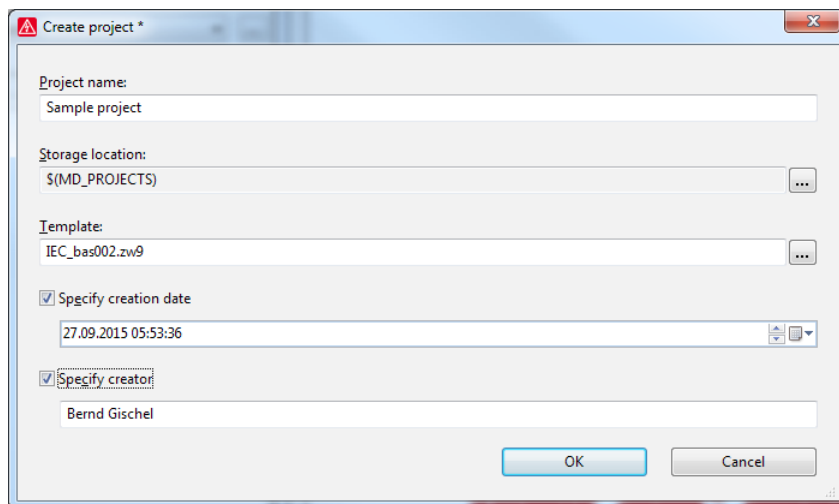


Fig. 14.3 Changed inputs in the Create project dialog

When you have entered all the information, you can close the dialog by clicking **OK**. EPLAN uses the inputs to create a new project.

After creating the project, EPLAN automatically opens the **PROJECT PROPERTIES** dialog where you can enter other information for the project, such as a project description or a commissioning number.



NOTE: It is usually possible to change project properties later on. This means that these project properties – more or less descriptive information – are not initially required for a project.

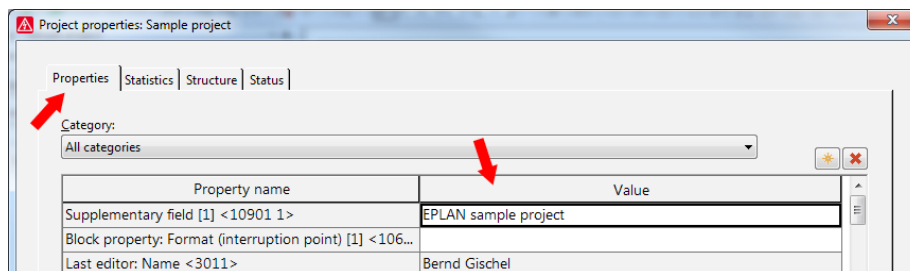


Fig. 14.4 Dialog for changing project properties

Now you can close the **PROJECT PROPERTIES** dialog. EPLAN then opens the new project in the **Page navigator**.

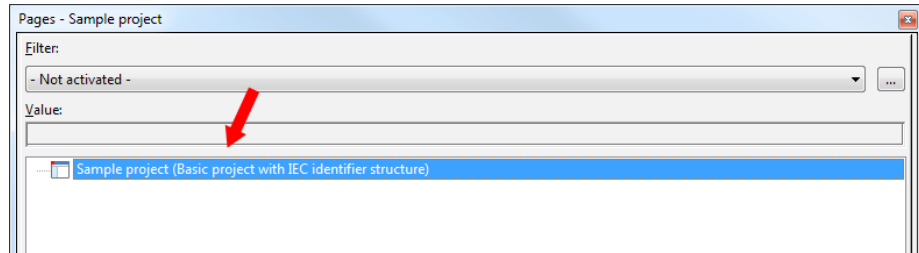


Fig. 14.5 Open project in the page navigator

■ 14.2 Generate pages

To proceed further with the project, for example to edit it in the graphical editor, EPLAN needs pages (maybe even different types of pages) that can be called up later in the **Page navigator**.

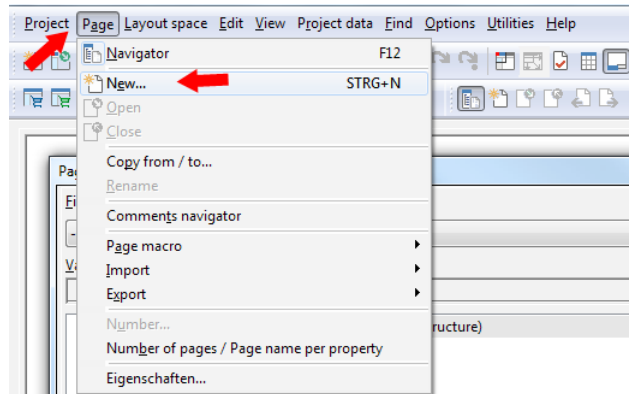


Fig. 14.6 Page menu

What are pages? Pages are needed in EPLAN, for instance, to create a graphical schematic. You can use the **PAGES/NEW** menu to create different types of pages.

What are page types? EPLAN uses different types of pages. A schematic would usually use pages that have the page type *Schematic multi-line* (Electrical engineering trade) or *Fluid power schematic* (Fluid power trade). A cover sheet would have the page type *Title page/cover sheet* and a report, e.g. a terminal diagram, would have the page type *Terminal diagram*. EPLAN makes a distinction between automatic pages that EPLAN automatically generates (such as a *terminal diagram*) and interactive pages that can be manually created and edited (such as a *multi-line schematic*).

What is a graphical editor? The graphical editor is the part of the EPLAN program that is used to create and change a schematic or overviews. The graphical editor contains other functions that can be accessed using various menus such as **INSERT** and **EDIT**.

What is a page navigator? The page navigator is the center of action for page editing. The page navigator can be used to create and rename pages, or to change and even delete their properties, as well as to filter them according to specific criteria.

Pages are created using the **PAGE/NEW** menu. Then you define the page properties (see Fig. 14.7 and Table 14.2).



NOTE: For a schematic project, you should always start with a cover sheet (page type *Title page / cover sheet*). The page type can be automatically generated, or (as an exception to the rule) manually created.

New page

Full page name: /1

Page type: Title page / cover sheet (A) <17>

Page description: Title page

Properties

Category: All categories

Property name	Value
Plot frame name <11016>	
Form name <11015>	F26_001
Scale <11048>	1:1
Grid <11051>	1,00 mm
Previous page <11041>	
Start value (row) <12026>	0
Frozen report page <11049>	☐
Supplementary field [1] <11901 1>	

Buttons: Clear fields, OK, Cancel, Apply

Fig. 14.7 Page properties dialog

Table 14.2 The most important inputs for a page

Setting	Comment	Example
Full page name	Specify the structure of the page here. You can also use the [...] button to select an entry.	“1” or “10”
Page type	Use the enter [...] button to select the page type (<i>Schematic multi-line</i> , <i>Title page/cover sheet</i> , etc.).	Title page /cover sheet“ or “Schematic multi-line”
Page description	Enter the description of the page here.	“Cover sheet”
<i>Optional</i> ; Properties	<i>Optional</i> ; form name (Note: is usually automatically filled in by EPLAN)	<i>Optional</i> ; F26_001 (here: special page type Cover sheet / Title page created manually)

When you have entered all the information, you can close the dialog by clicking **OK**. EPLAN using the inputs to generate a new page and displays it in the page navigator.

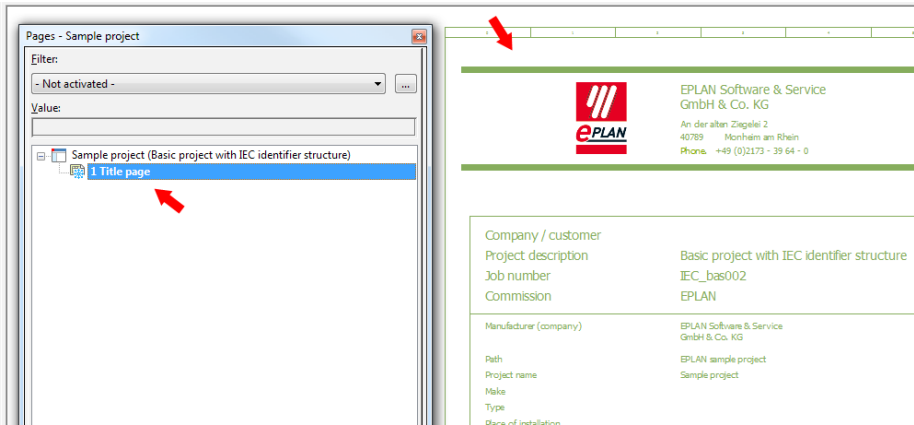


Fig. 14.8 First page — cover sheet

You use the same steps to create the next page.

- Call using the **PAGE/NEW** menu.
- Enter the **Full page name** in the **NEW PAGE** dialog.
- Select the **Page type** in the **NEW PAGE** dialog (Note: Many page properties, including the **Page type**, are taken from the currently selected page so you might have to use the selection list to change the page type.)
- Enter the **Page description** in the **NEW PAGE** dialog.
- Click **OK** to save and close the **NEW PAGE** dialog.

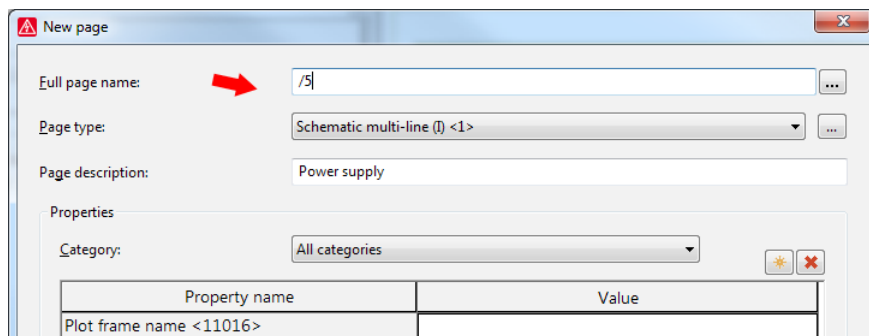


Fig. 14.9 New page with page type Schematic multi-line

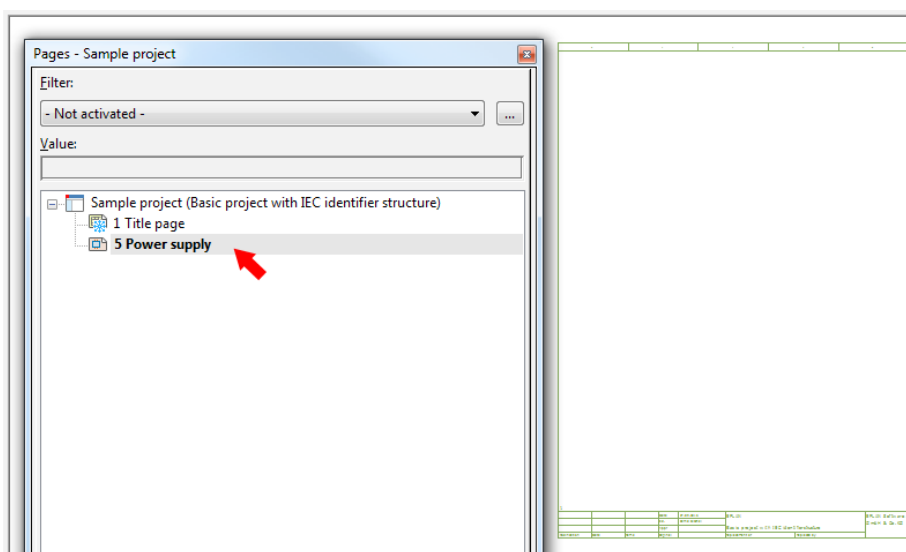


Fig. 14.10 Finished page in the page navigator



TIP: To edit the page properties, select the page in the open **Page navigator** and right click to open the **PROPERTIES**.

When you have finished creating all the pages you need, you can edit them. In the **page navigator**, double click the page you want to edit. EPLAN opens the page in the graphical editor.

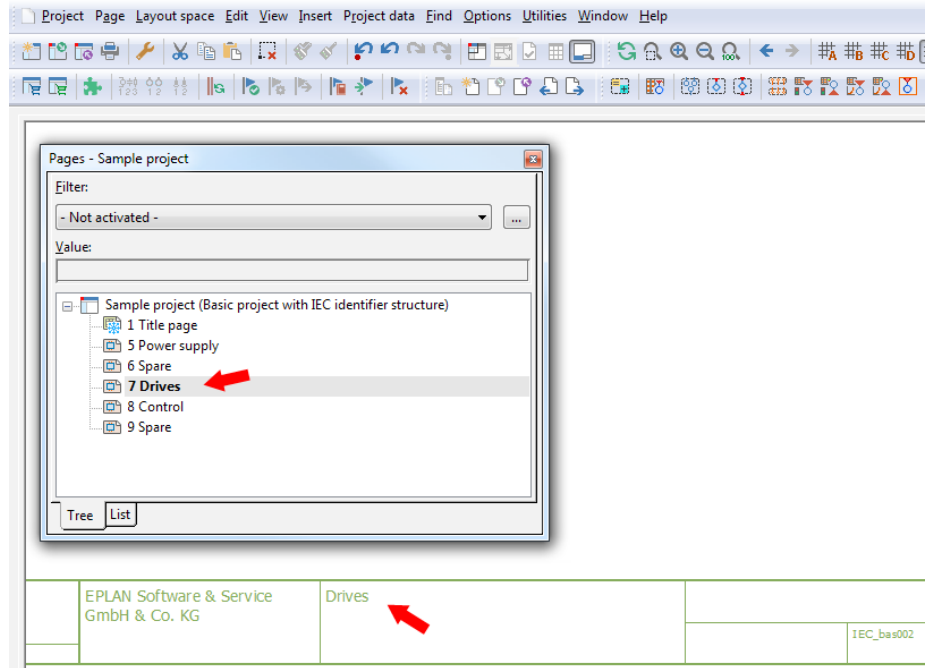


Fig. 14.11 Open page in the graphical editor

■ 14.3 Create a schematic

A schematic mainly consists of various symbols, components, graphical elements and usually so-called connectors that are used, for example, to automatically implement auto-connecting connections.

What is a symbol? A symbol is a graphical representation that EPLAN uses to represent functions. A function can be a motor, a coil, a signal lamp, etc. A symbol is initially only a graphic or a graphical representation and does not possess any logical data.

What is a component? A component is a symbol that has an associated logical function. The function itself contains logical data, such as other function definitions. A coil for example, contains functions other than just the coil function, e.g. two NO contacts and two NC contacts. A component is the combination of a graphical symbol (the coil) and its functions.

What are graphical elements? Graphical elements can be lines, circles and even rectangles. Graphical elements have no logical functions and are simply drawing elements that have no impact on the logical content of a schematic.

What are connectors? Connectors are special symbols that allow you to use existing automatic connects to “forward” logical data or to simply create a logical connection in the first place. Connectors include angles, T-nodes and interruption points.

What are autoconnection connections? One advantage of EPLAN is the so-called autoconnection connection. As soon as two connection points for a symbol or connector are precisely opposite one another, they are automatically (= autoconnecting) connected by a type of “line”. EPLAN recognizes and outputs only these autoconnecting lines as logical connections. Connections that are simply a graphical line drawn between two connection points of a symbol are not logical connections.

14.3.1 Insert symbols

To insert a symbol, you press the CTRL key. EPLAN opens the SYMBOL SELECTION dialog.

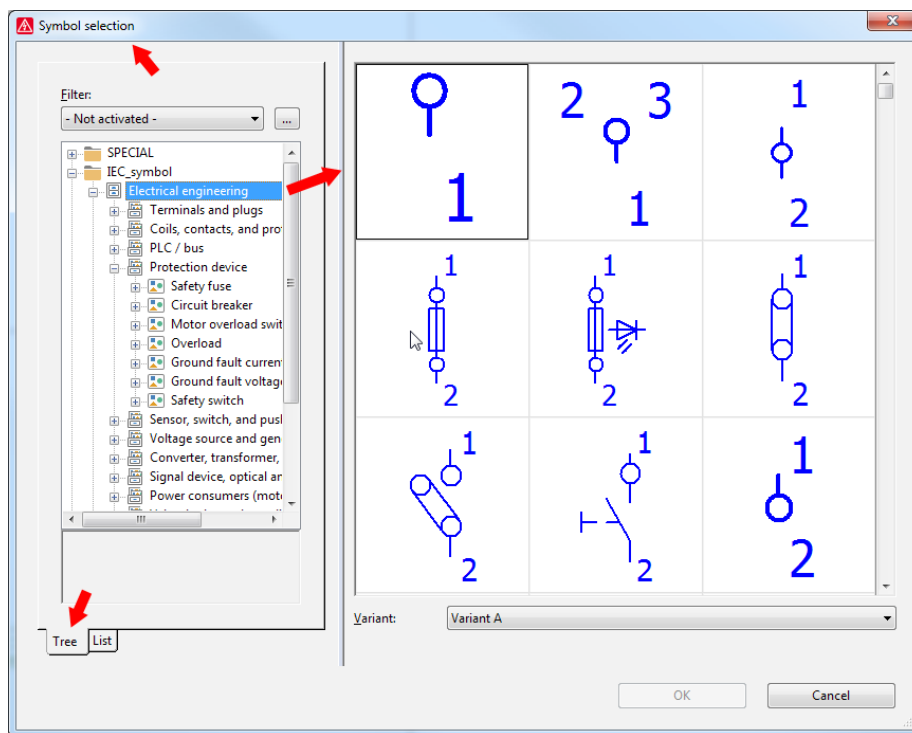


Fig. 14.12 Symbol selection dialog

In this dialog, you select the appropriate symbol and suitable variant (rotation angle) and click OK to apply it. EPLAN “hangs” the symbol on the cursor, and it can then be placed anywhere on the page. If the wrong variant was selected, the symbol can be rotated using the TAB key before being placed (the variant is exchanged).

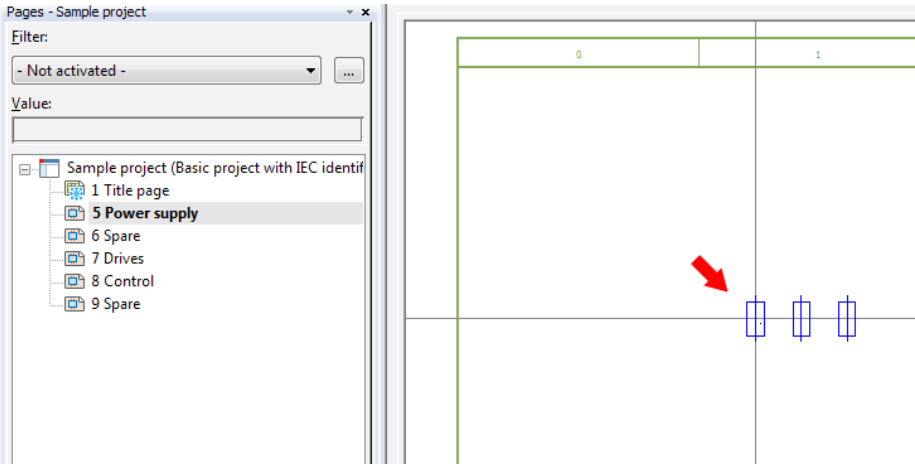


Fig. 14.13 Placing a symbol

After placement, EPLAN opens the **PROPERTIES (COMPONENTS) GENERAL DEVICE** dialog. Some of the properties may have default values. For example, the device tag (DT) if online numbering was activated in the project settings.

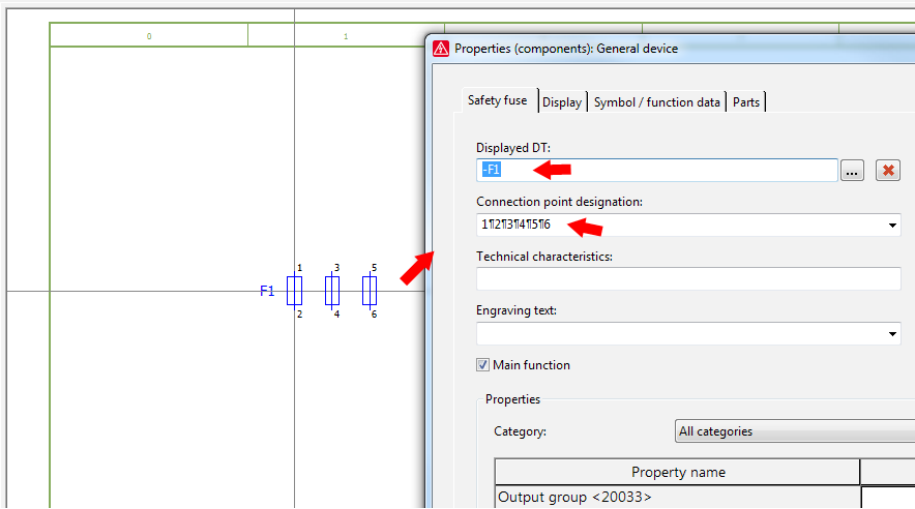


Fig. 14.14 Properties (component): General devices dialog

At a minimum, the following entries must be entered for the symbol in the dialog:

Setting	Comment	Example
Displayed DT	The actual device tag	F1
Connection point designations	Designations for the connection points; for a coil this could be f A1 A2	1 2 3 4 5 6 Note: You can enter the “ ” separator by pressing CTRL + ENTER.
<i>Optional</i> ; Technical characteristic	<i>Optional</i> ; Technical characteristic of the device	<i>Optional</i> ; 32A
Main function	Each DT can only have one main function. At least one must have a main function.	Click the box next to Main function to activate it



NOTE: EPLAN automatically creates the (grayed out) *Full DT* field from the page identifier, including the displayed DT. You usually do not have to enter this manually.

When you have entered all the information, you can save and close the dialog by clicking OK.

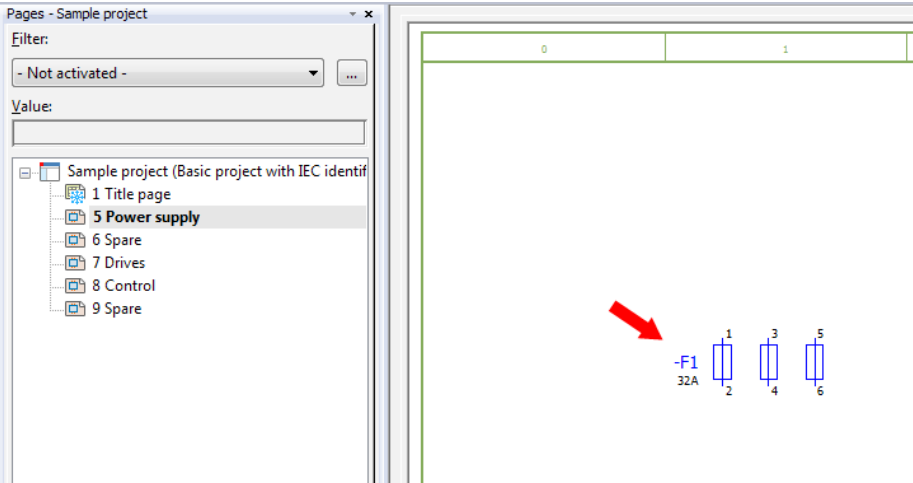


Fig. 14.15 Finished symbol with entered information

Now you can insert and place other symbols (terminals in the example). When you are done, you can use connectors to connect them. For example, you could use the **INSERT/CONNECTOR** to select and place an angle.

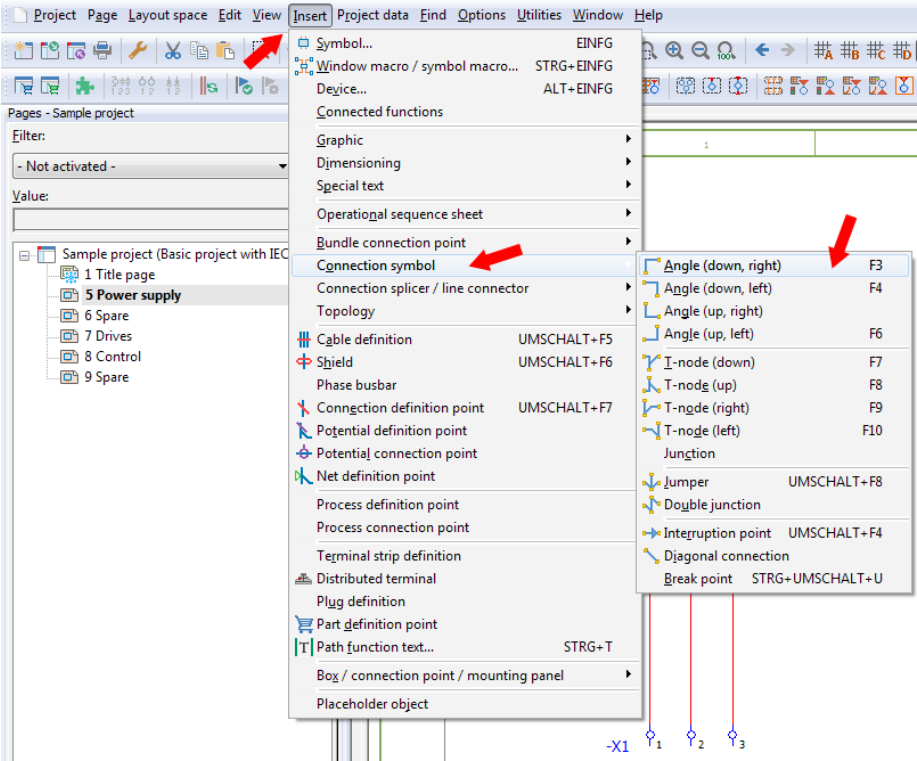


Fig. 14.16
Placing connection
symbol

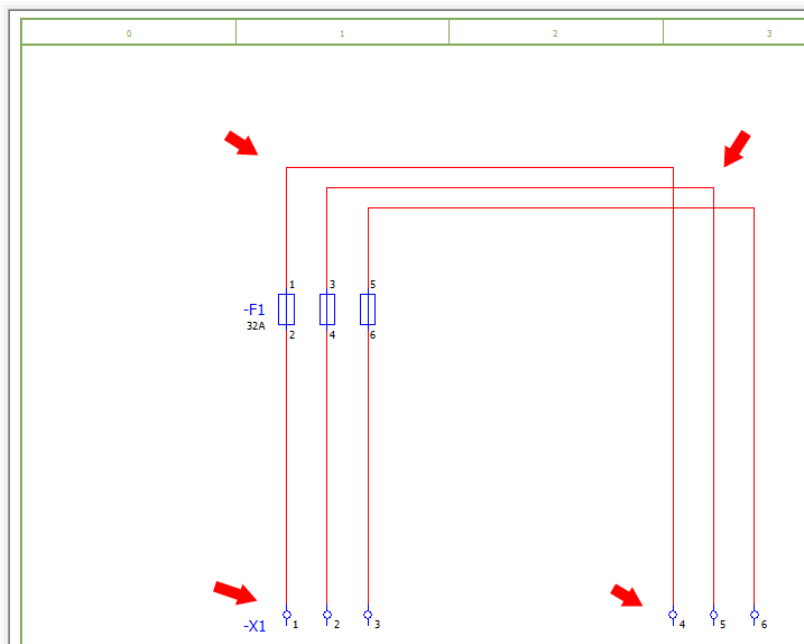


Fig. 14.17
Schematic with
connection symbols

The terminals in the figures are inserting like normal devices using the **INSERT/SYMBOL SELECTION** menu or the **CTRL** key.

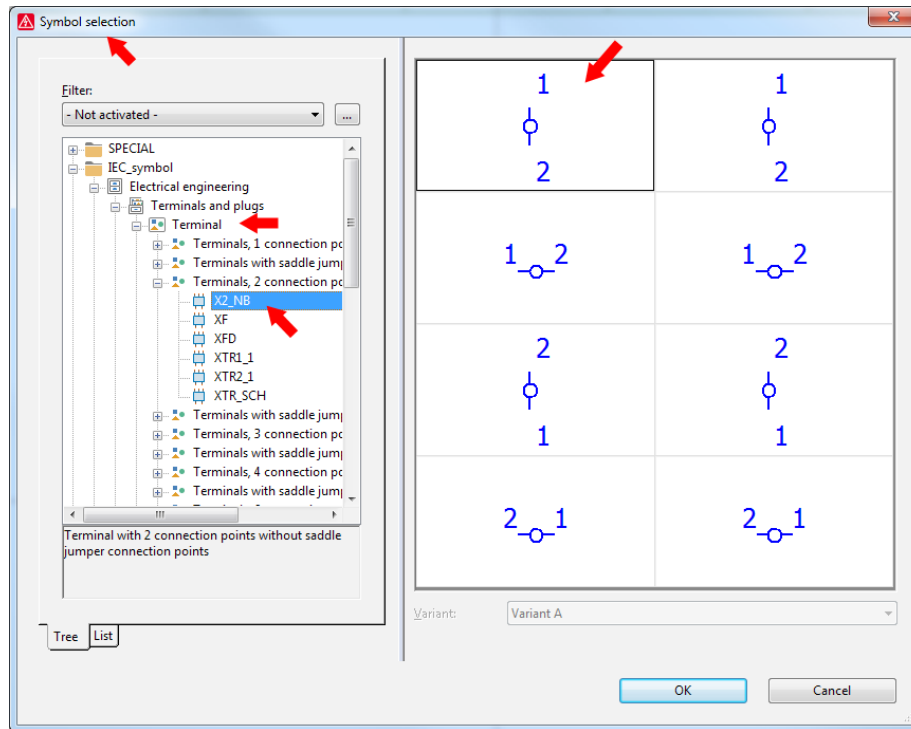


Fig. 14.18 Inserting terminals

After placement, EPLAN opens the **PROPERTIES (COMPONENTS): TERMINAL** dialog. In this dialog, you enter the **Displayed DT** and the **Connection point designation** (terminal or plug designation).

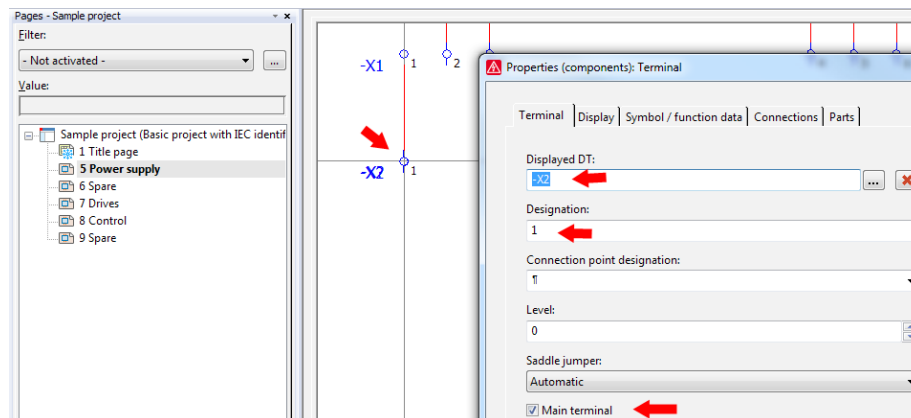


Fig. 14.19 Properties (component): Terminal dialog

14.3.2 Insert cables

In addition to devices like coils, terminals, etc., you can also insert cables. You place cables using **SHIFT + F5** or the **INSERT/CABLE DEFINITION** menu.

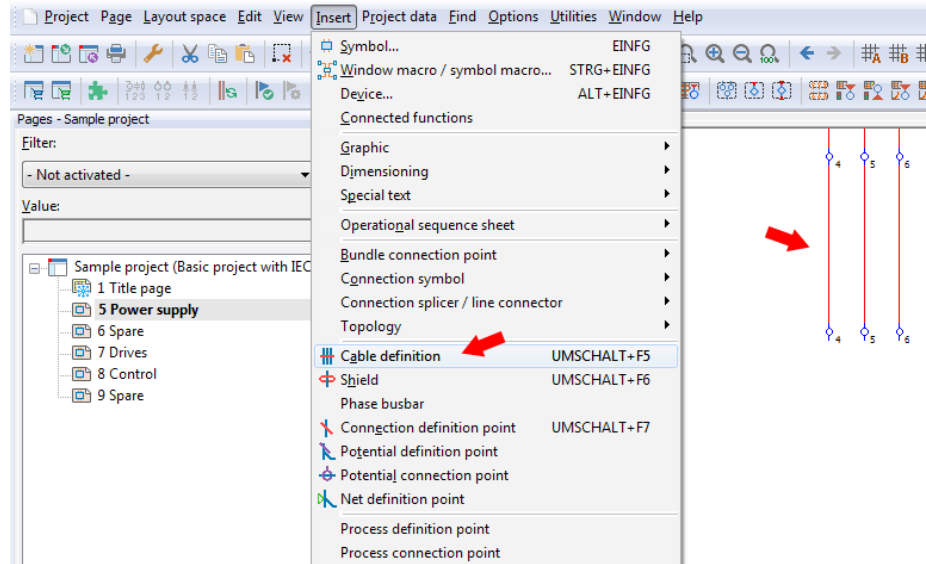


Fig. 14.20 Inserting cables

When a cable is selected, the cable definition “hangs” on the cursor and you can now insert the cable between devices. You left click to select the starting point (usually on the left).

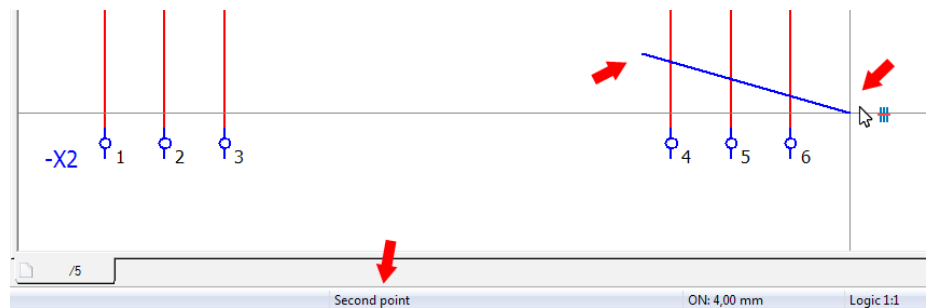


Fig. 14.21 Starting point of the cable definition

Now you drag the cable via the connection and left click to set the second point. EPLAN then opens the **PROPERTIES (COMPONENT): CABLE** dialog where you enter data such as the **Displayed DT**.

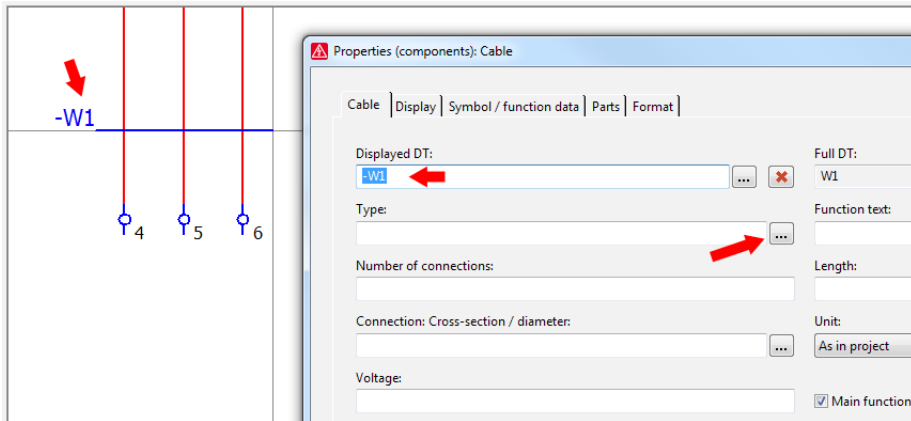


Fig. 14.22 Properties (component): Cable dialog

You can now add other cable data, such as **cable type** from **parts management**. The best way is to click the [...] button and select and apply the cable from parts management.

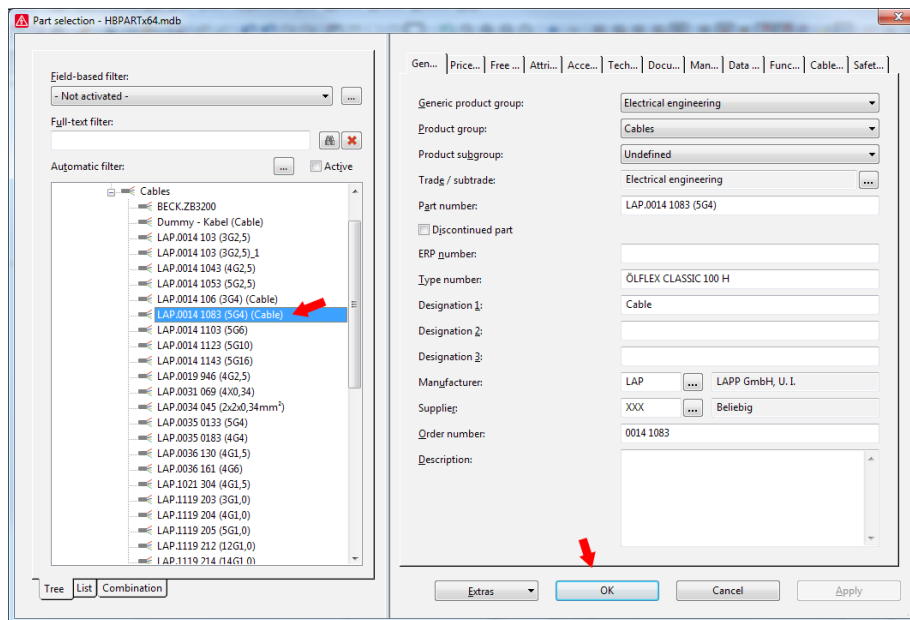


Fig. 14.23 Selecting a cable from parts management

When you click **OK**, parts management closes and EPLAN automatically adds the entered cable data such as cable type, number of conductors, etc.

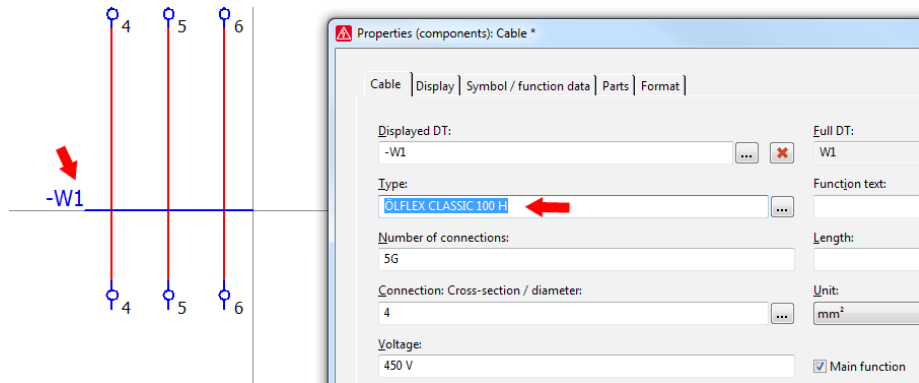


Fig. 14.24 Added cable data

When you click **OK** in the **PROPERTIES (COMPONENTS): CABLE** dialog, EPLAN closes the dialog and displays the cable and all its data in the schematic.



TIP: You can also manually enter, without entering a part selection, additional cable data, such as cable type, number of conductors and conductor color.

To label the conductors, select the cable and then the **PROJECT DATA/CABLE/ASSIGN CONDUCTORS REASSIGN ALL** menu item. EPLAN automatically assigns the conductors to the cable and labels them.

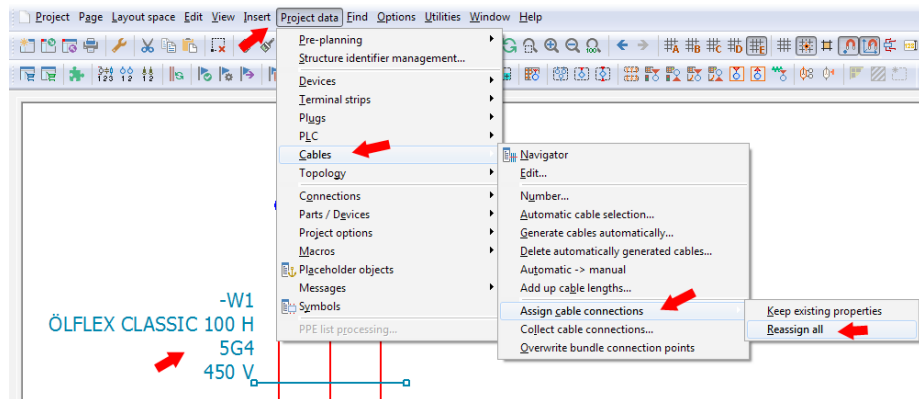


Fig. 14.25 Assigning conductors to the cable